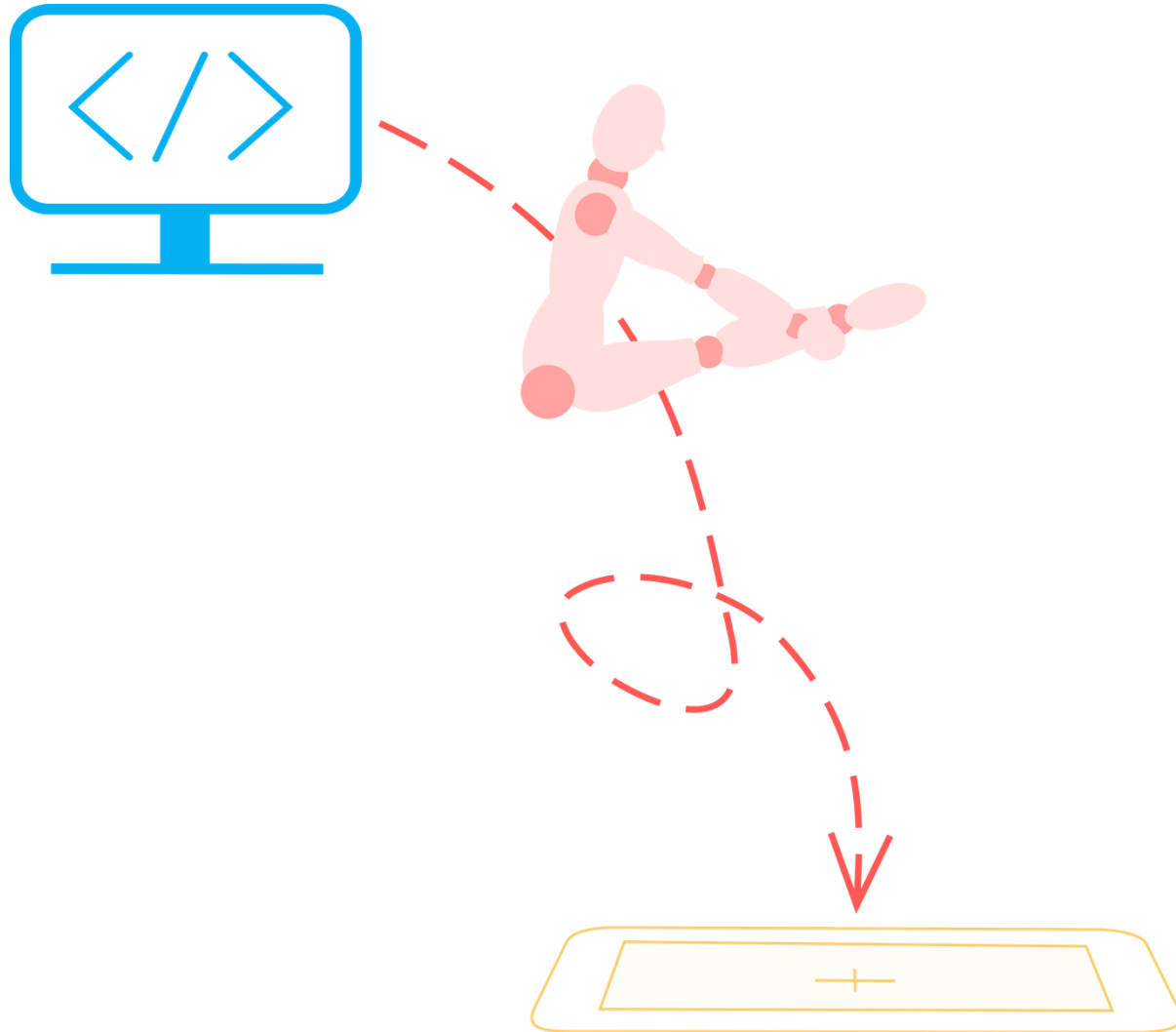


Jumping out of the computer onto the field:

Some coaching principles for acrobatic sports



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Annie Ross

Polytechnique Montréal

Mickaël Begon

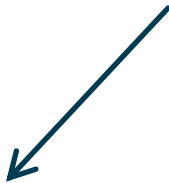
Université de Montréal

Overview

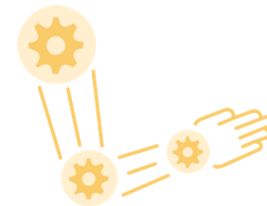
- Methods (quickly)
- Principles:
 - Biomechanics (5)
 - Perceptual / vision (1)
 - Motor control (1)
- Tools (quickly)
- Group discussion



Trampolinist and coach



End of PhD



Biomechanist

Predictive simulations



Dynamics

- $\tau = M(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q)$

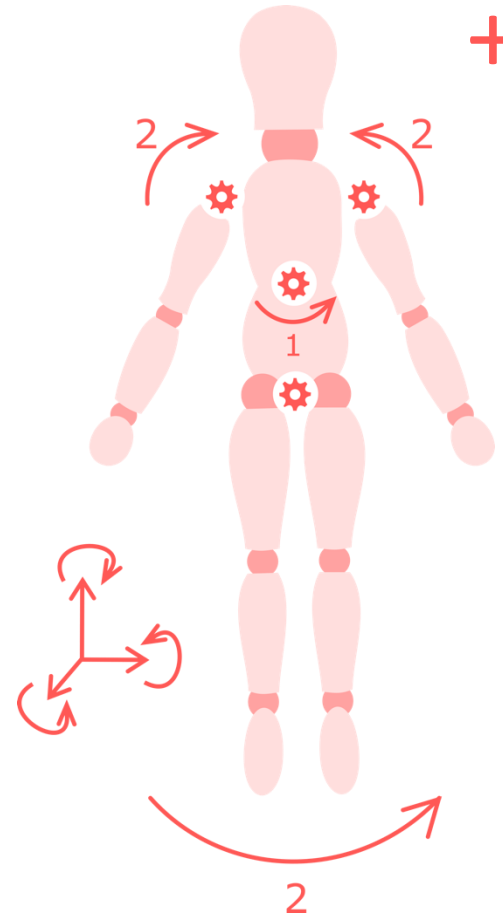


Performance objectives

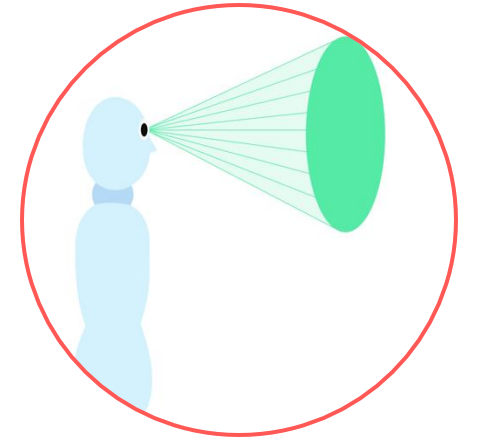


Constraints

- Acrobatics
- Physiological



+

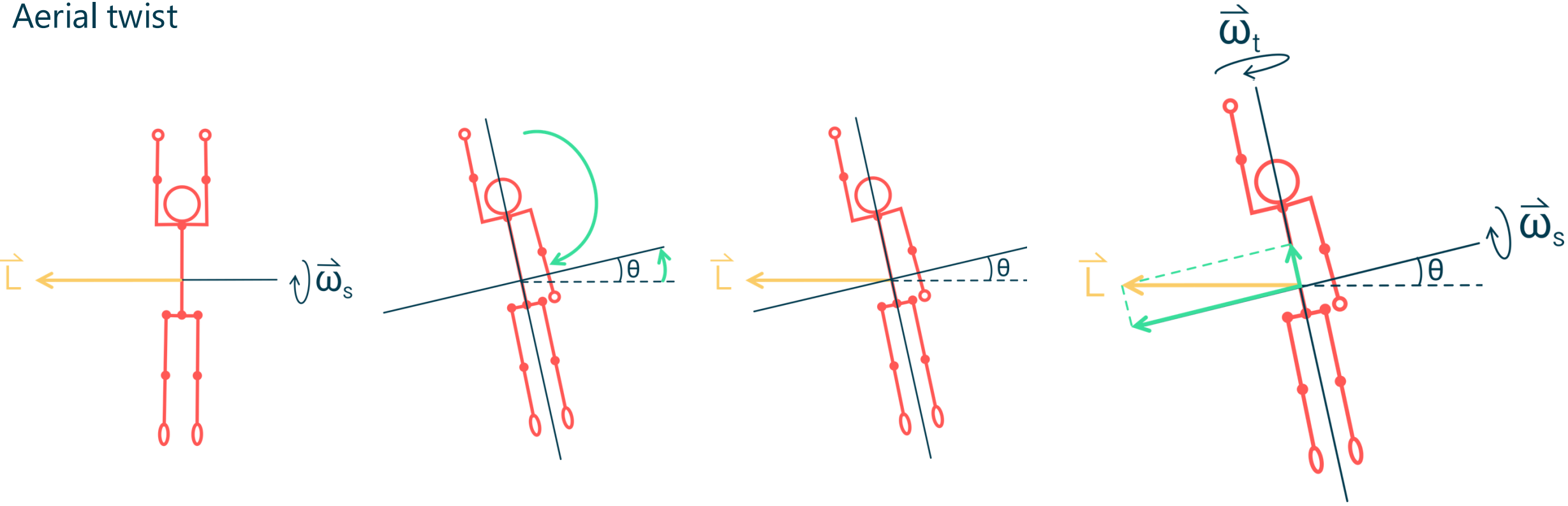


- 1 Moving segments in the best tilting plane
- 2 Circular motion of the hips
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- 6 Looking at the trampoline without fixating
- 7 Vision helps regulating variability

1

Moving segments in the best tilting plane

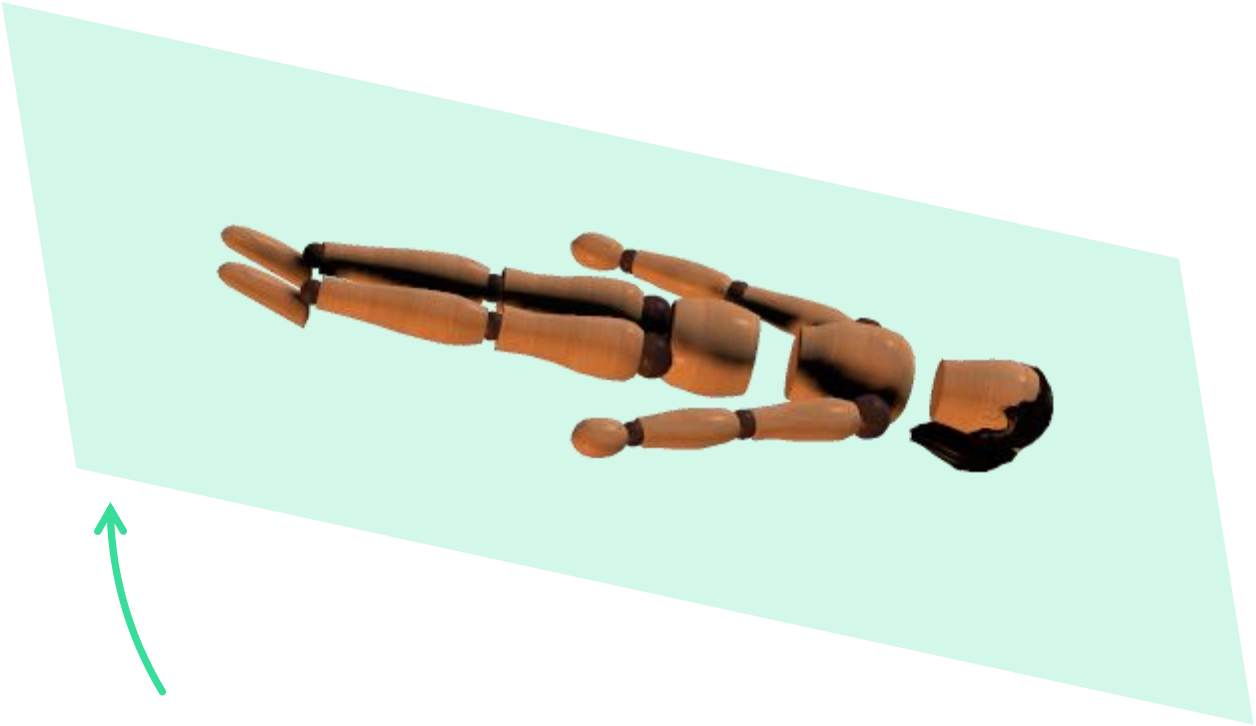
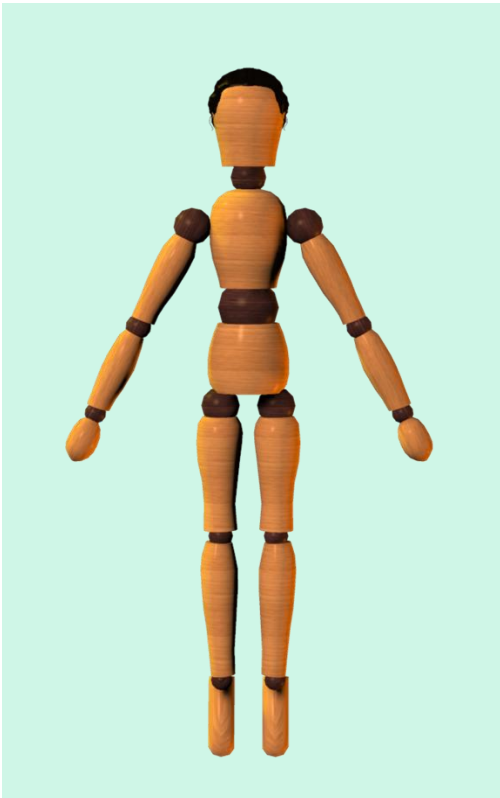
Aerial twist



1

Moving segments in the best tilting plane

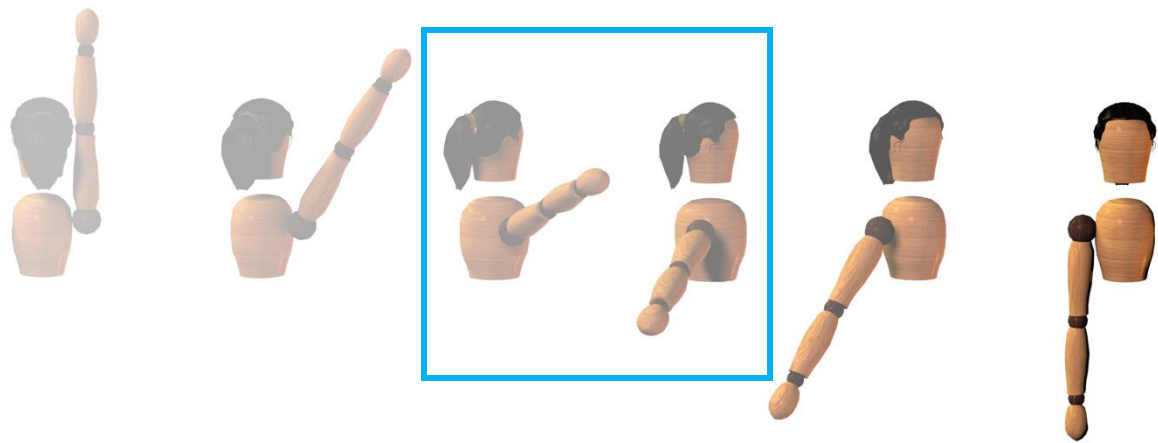
Best tilting plane



1

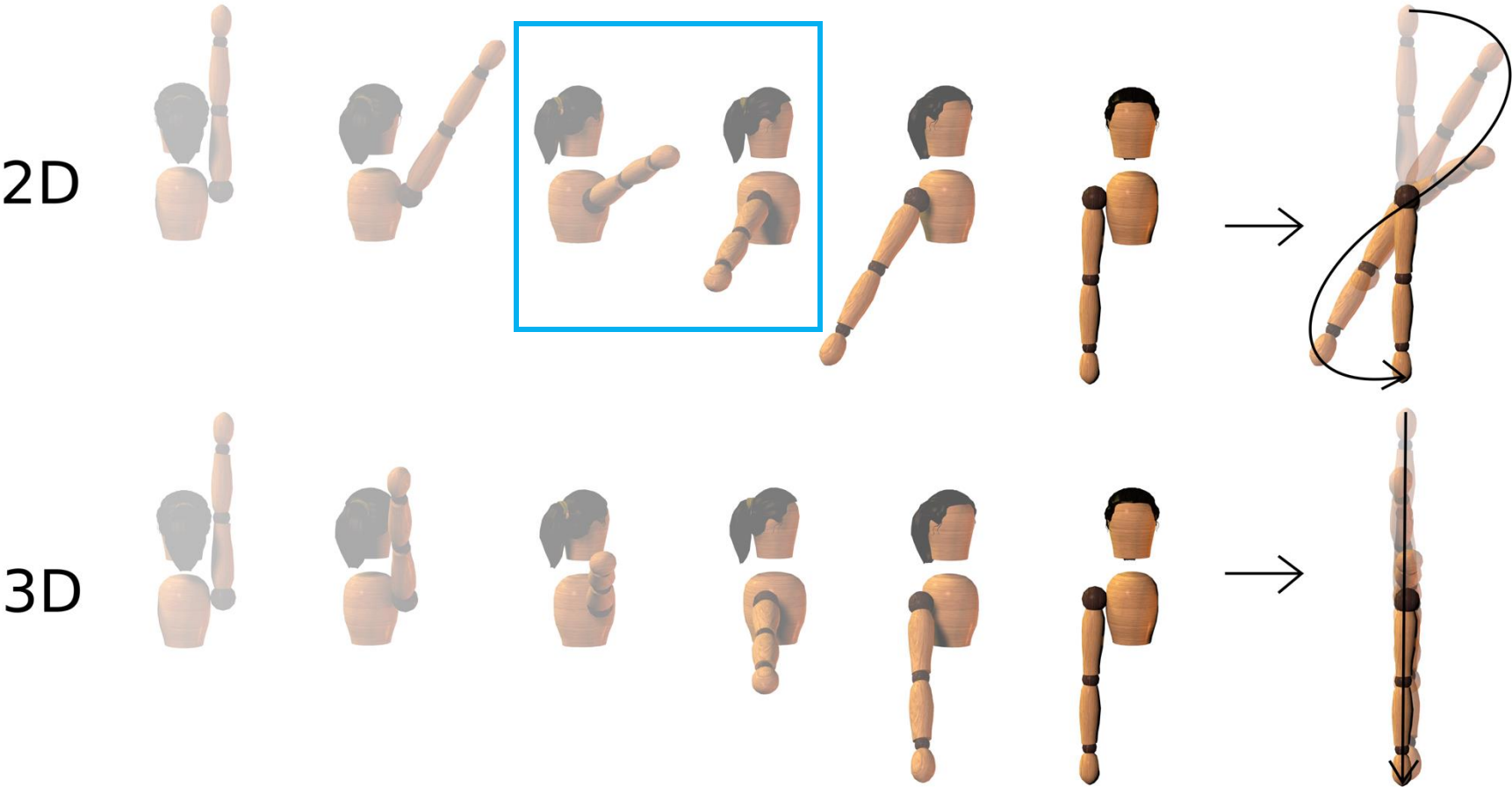
Moving segments in the best tilting plane

2D



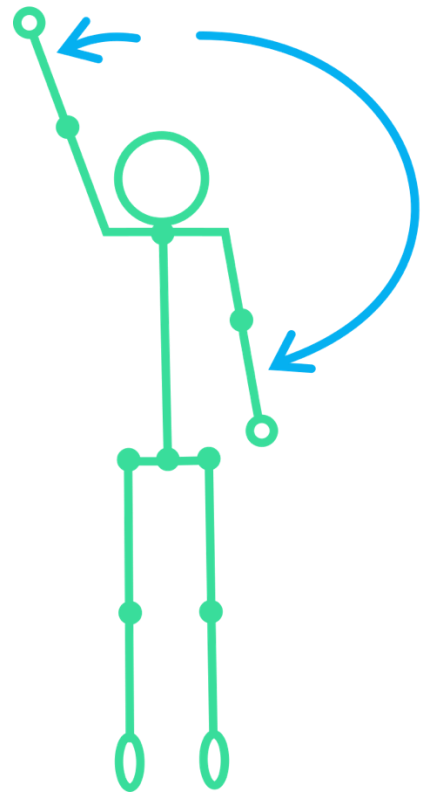
1

Moving segments in the best tilting plane

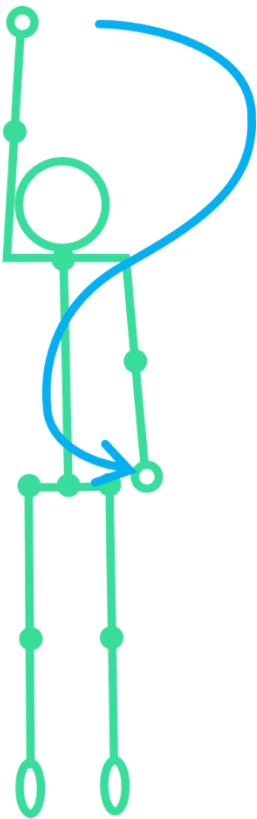


1

Moving segments in the best tilting plane



Coach instructions



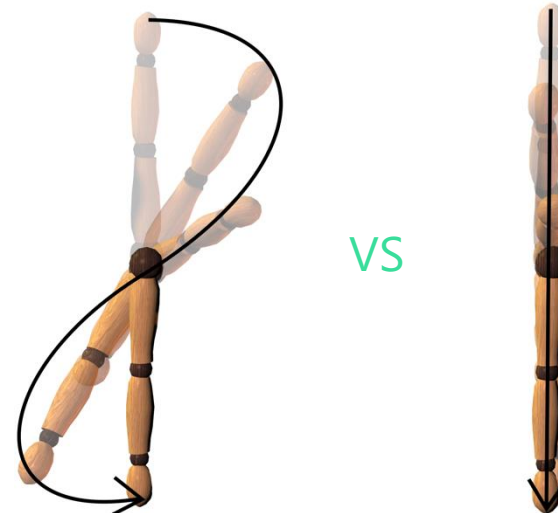
Athlete modifications

1 Moving segments in the best tilting plane



Conclusion

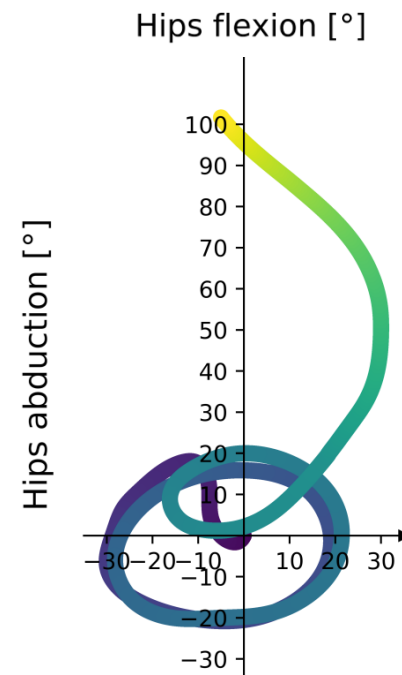
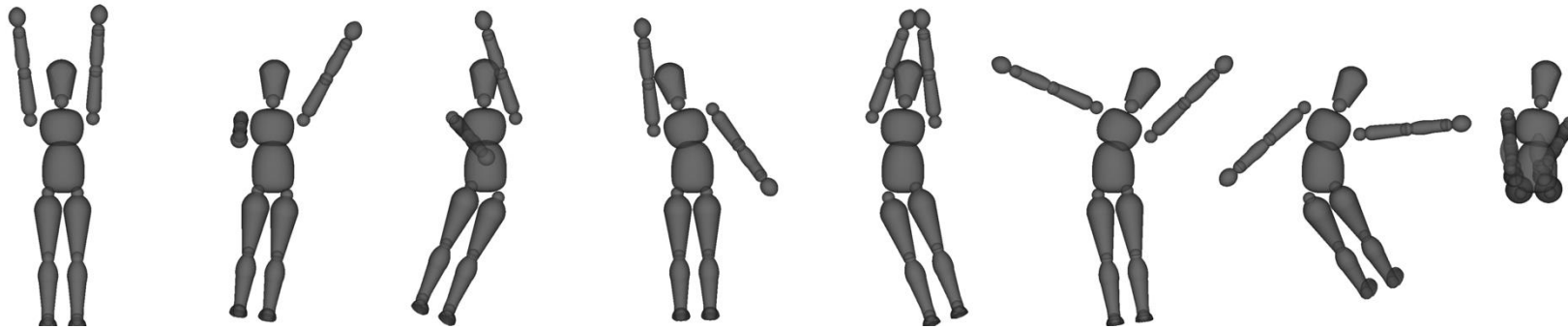
- ① Do not systematically correct for out of frontal plane movements
 - They are more performant
 - They are less visible!



- 1 Moving segments in the best tilting plane
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2

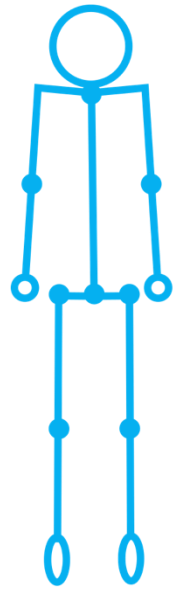
Circular motion of the hips



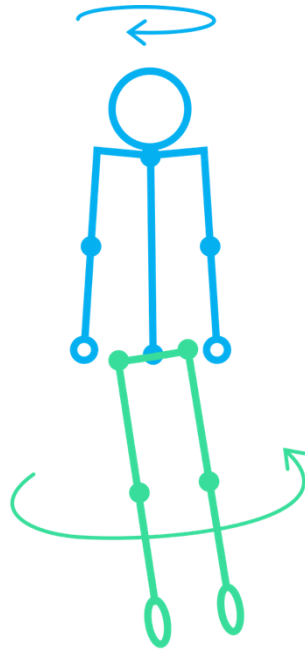
2

Circular motion of the hips

Cat twist



$$\vec{L} = 0$$



$$\vec{L} = I\omega + I\omega = 0$$

+ aerial twist contributions



2

Circular motion of the hips

Conclusion

- ② Teach circular motion of the hips
 - Early twists
 - Late twists
 - Untilting



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3

Adjustments might be needed before piking



① Twisting phase

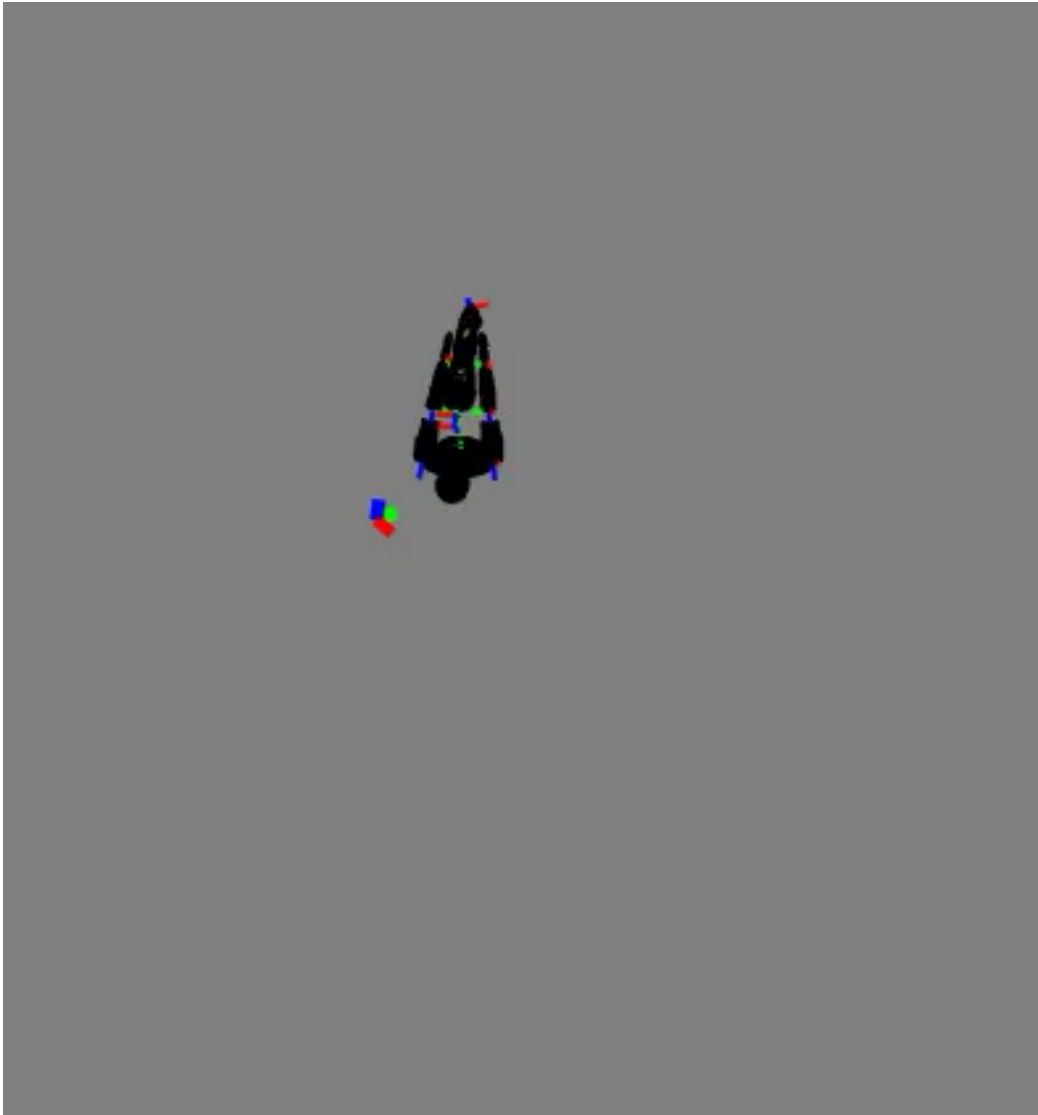
- Rotation: somersault & twist
- Oscillations: tilt

② Somersaulting phase

- Rotation: somersault
- Oscillations: tilt & twist

3

Adjustments might be needed before piking



Why is that problematic ?

- Not in line with sport regulations
- Harder to keep track of their orientation

How to transfer from twist to somersault ?

- Rotation velocities
- Relative moment of inertia

$$\cos^2 \alpha_0 = A(B - C)/B(A - C)$$

3

Adjustments might be needed



Already exists in other sports

Before



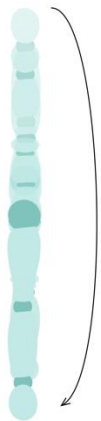
After



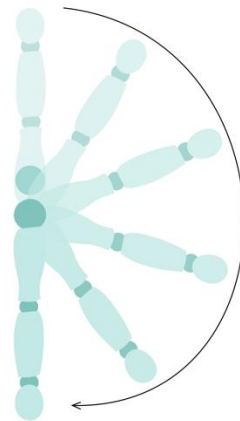
3 Adjustments might be needed

Conclusion

- ③ Do not push athletes to rush into the pike
 - Might lead to instability
 - Looser pike position
- ④ Help athletes make these adjustments
 - Hide the movements from the judges (in the BTP)



Least noticeable



Most noticeable

Before



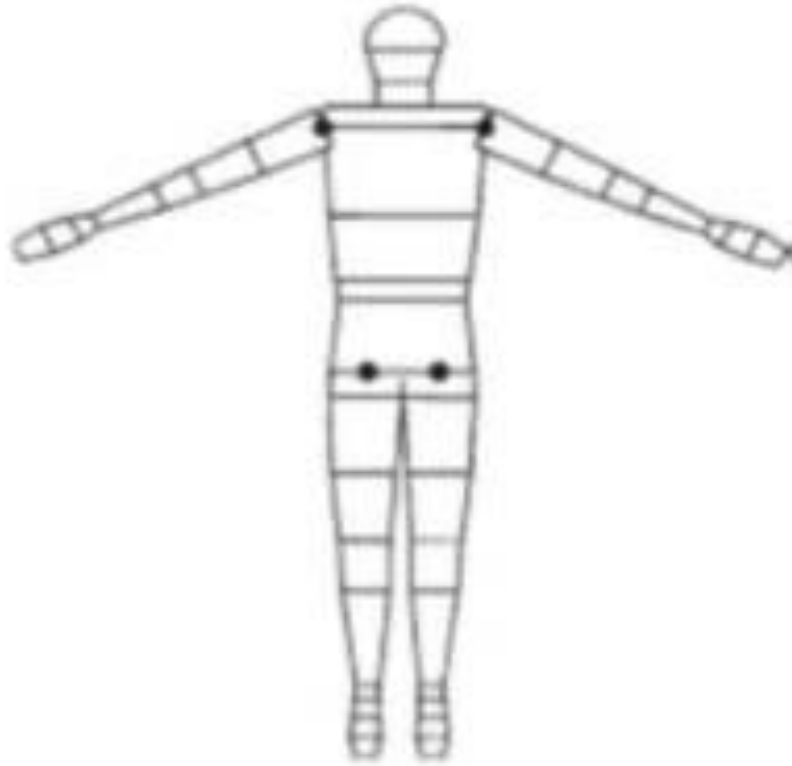
After



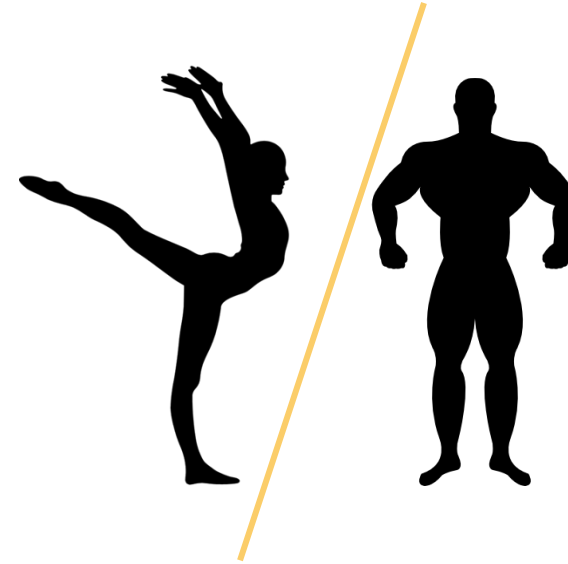
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4

Techniques are dependent on the athlete's anthropometry



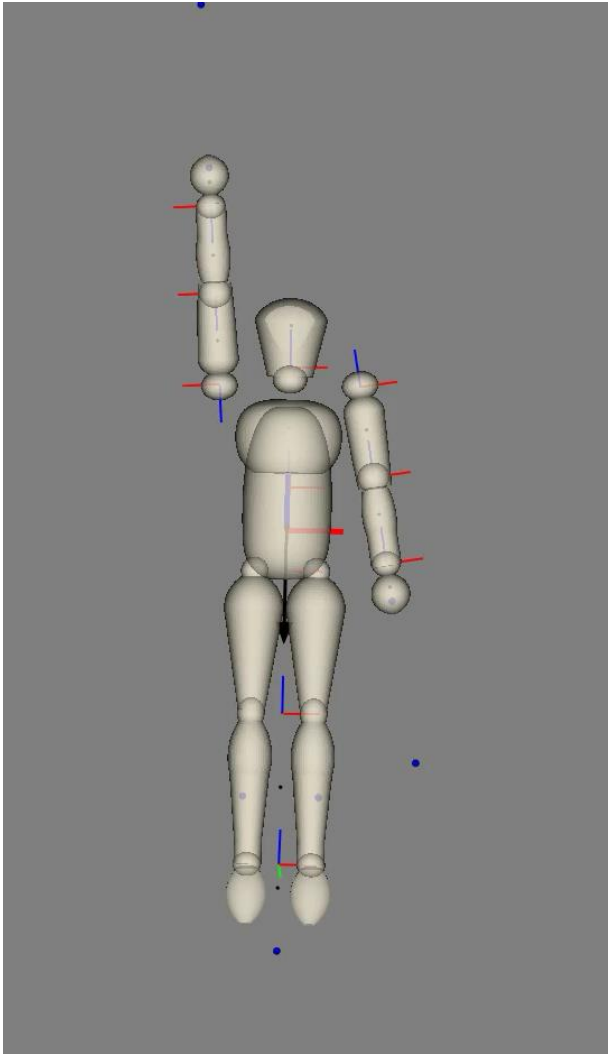
Yeadon's model



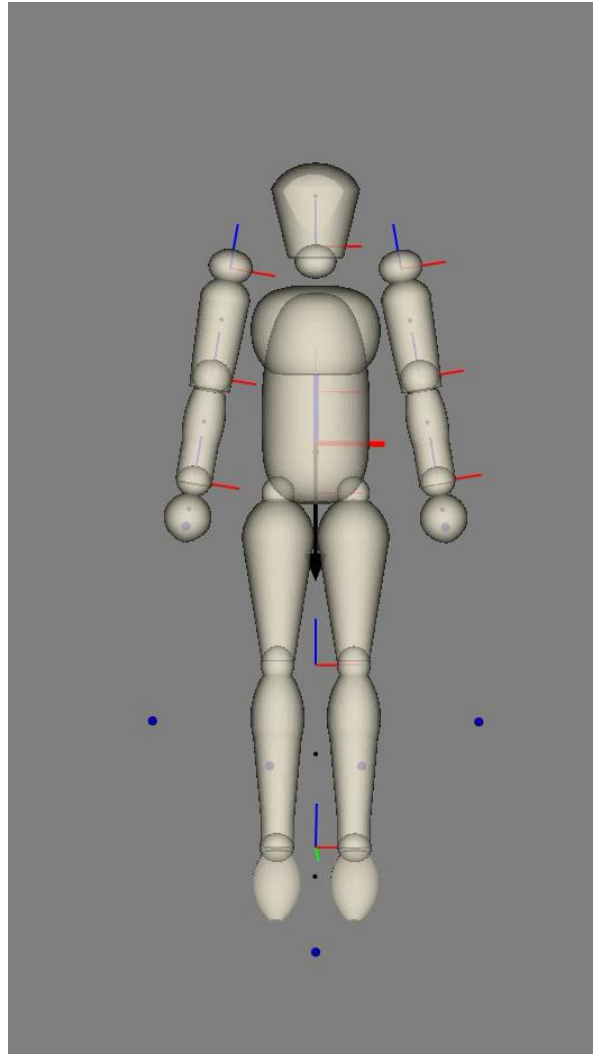
- 4 male gymnasts
- 4 female gymnasts
- 3 male trampolinists
- 2 female trampolinists
- 2 male divers
- 2 female divers
- 1 female artistic swimmer (flyer)

4

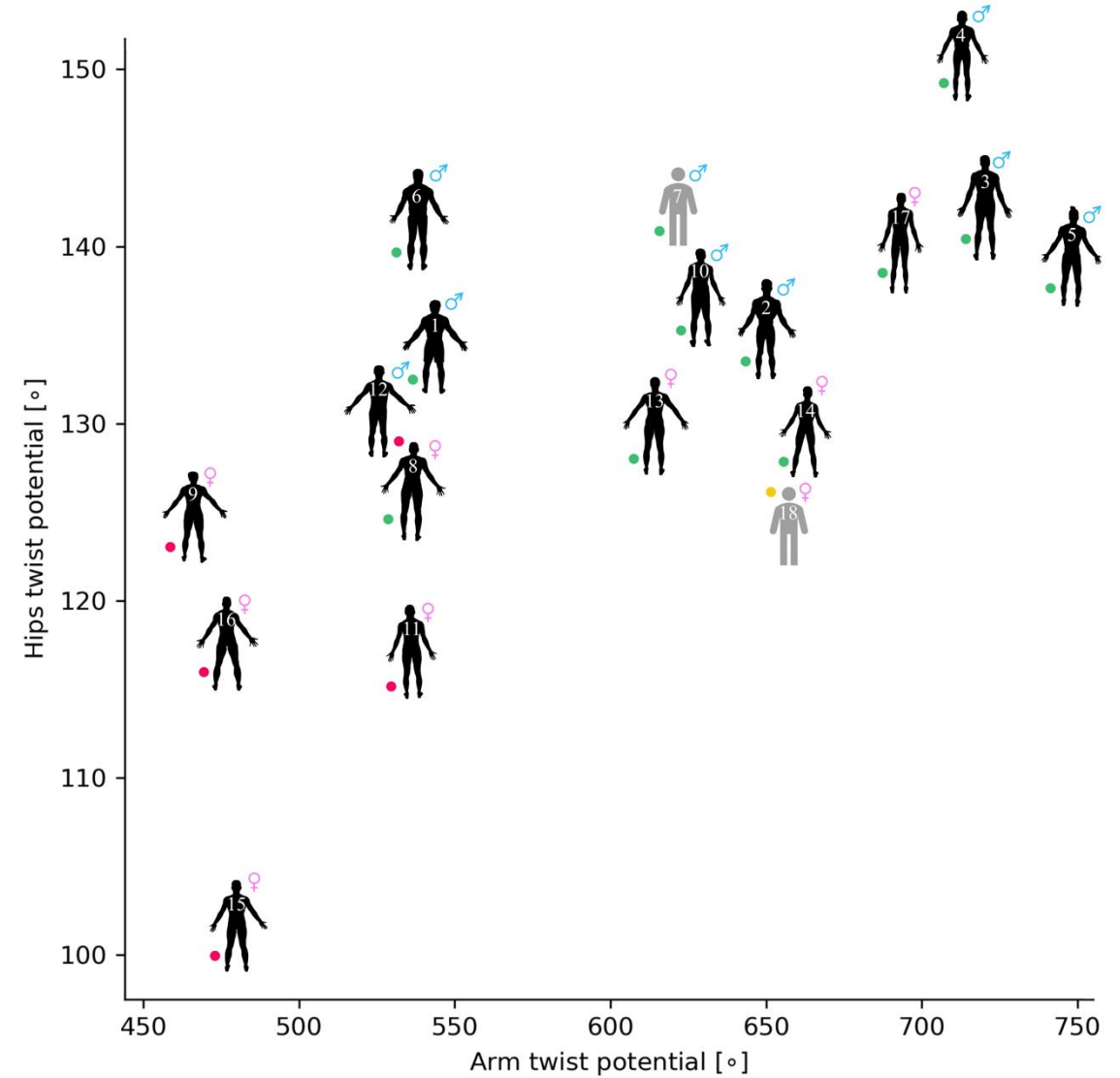
Techniques are dependent on the athlete's anthropometry



Arms twist potential

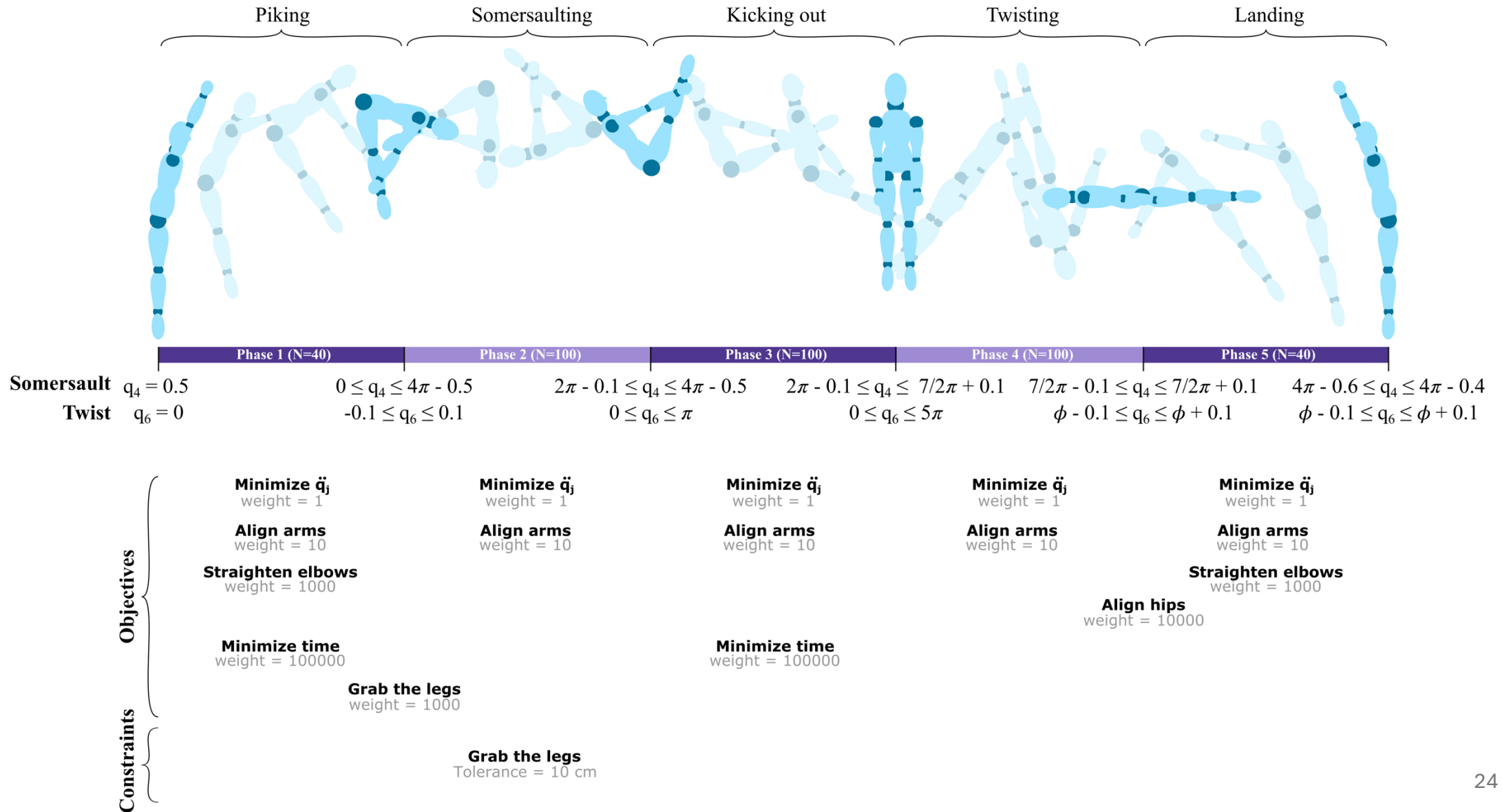


Hips twist potential



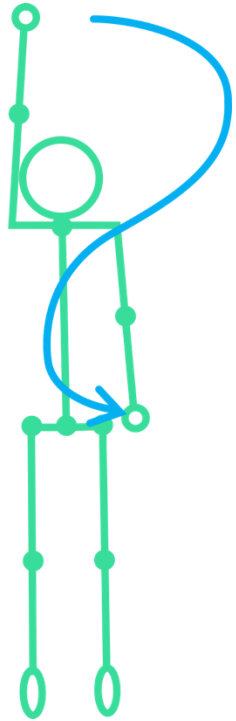
4

Techniques are dependent on the athlete's anthropometry

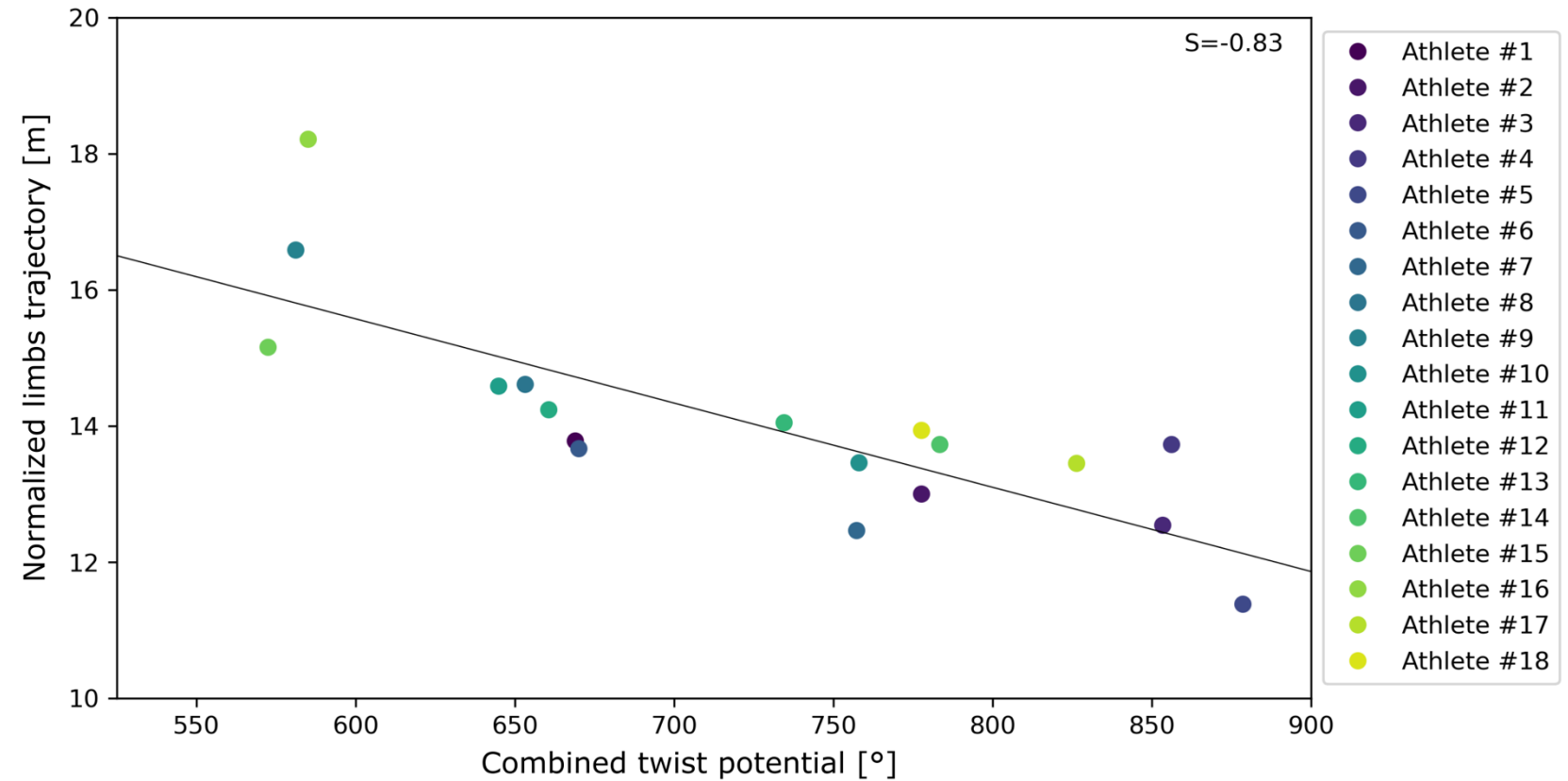


4

Techniques are dependent on the athlete's anthropometry



Normalized limbs trajectory

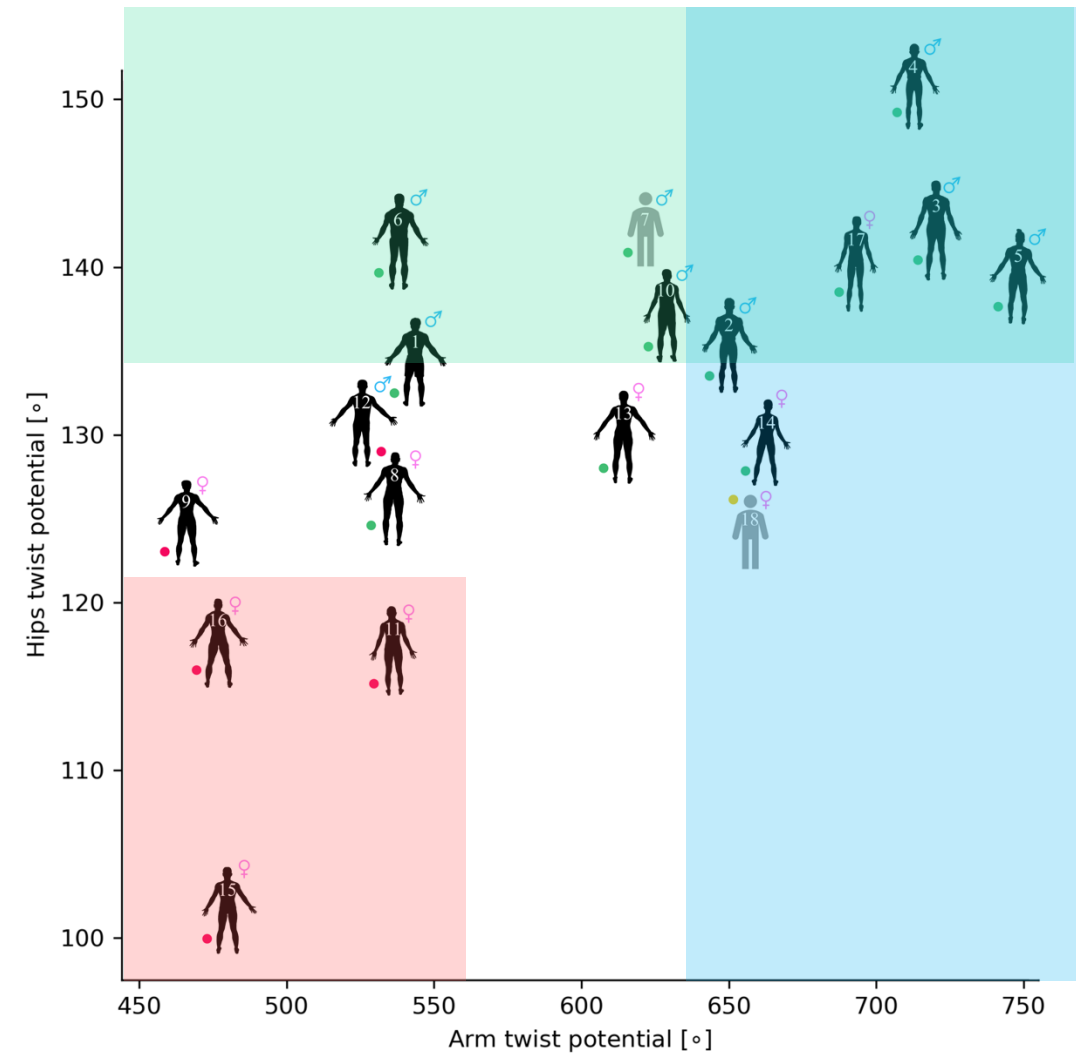
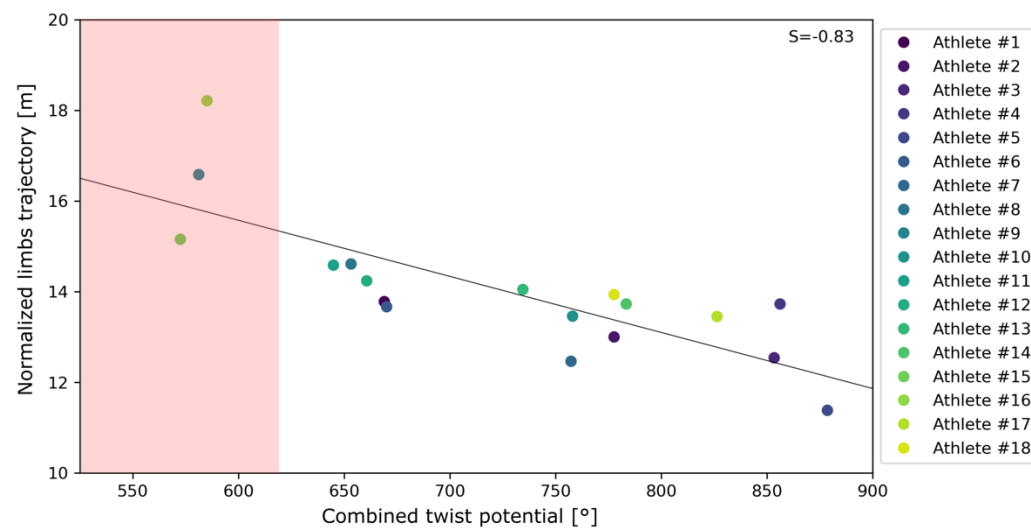


4

Techniques are dependent on the athlete's anthropometry

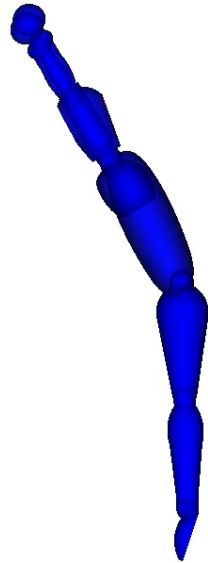
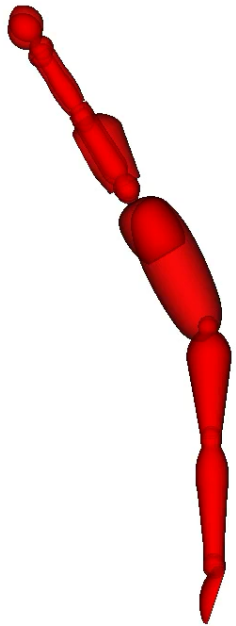
Conclusion

- ⑤ The twist potential can be used to choose which skill/technique is more appropriate
- Favor somersaulting skills for **low TP**
 - Be careful of **women** TP (small sample!)
 - Favor hip techniques for high HTP
 - Favor arm techniques for high ATP

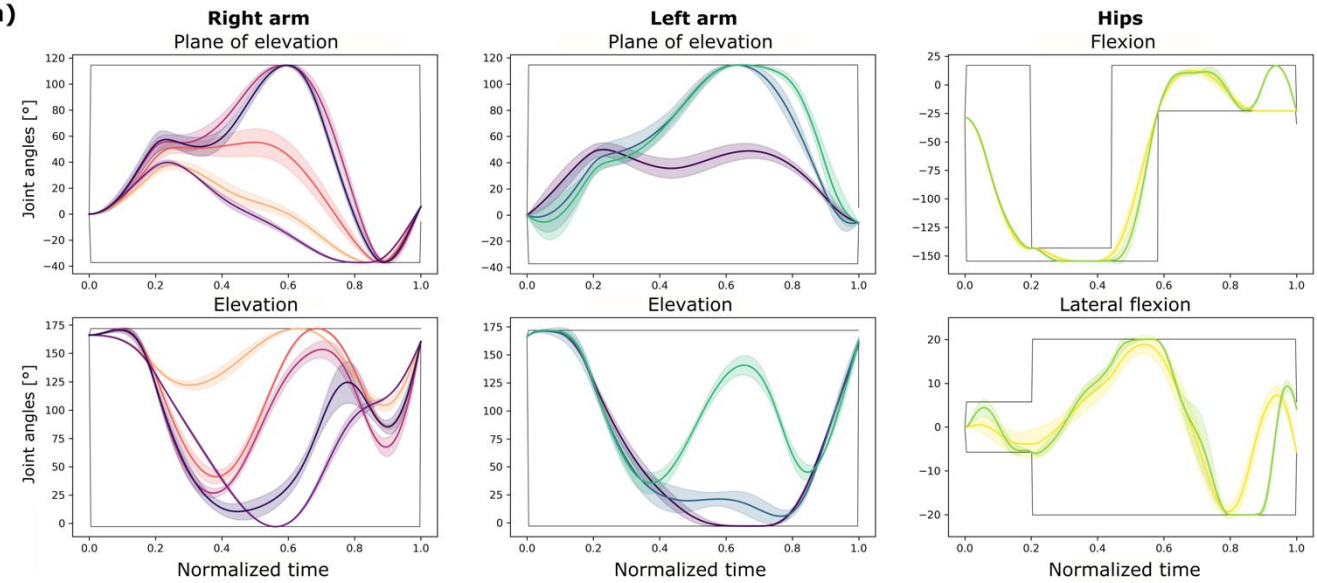


4

Techniques are dependent on the athlete's anthropometry



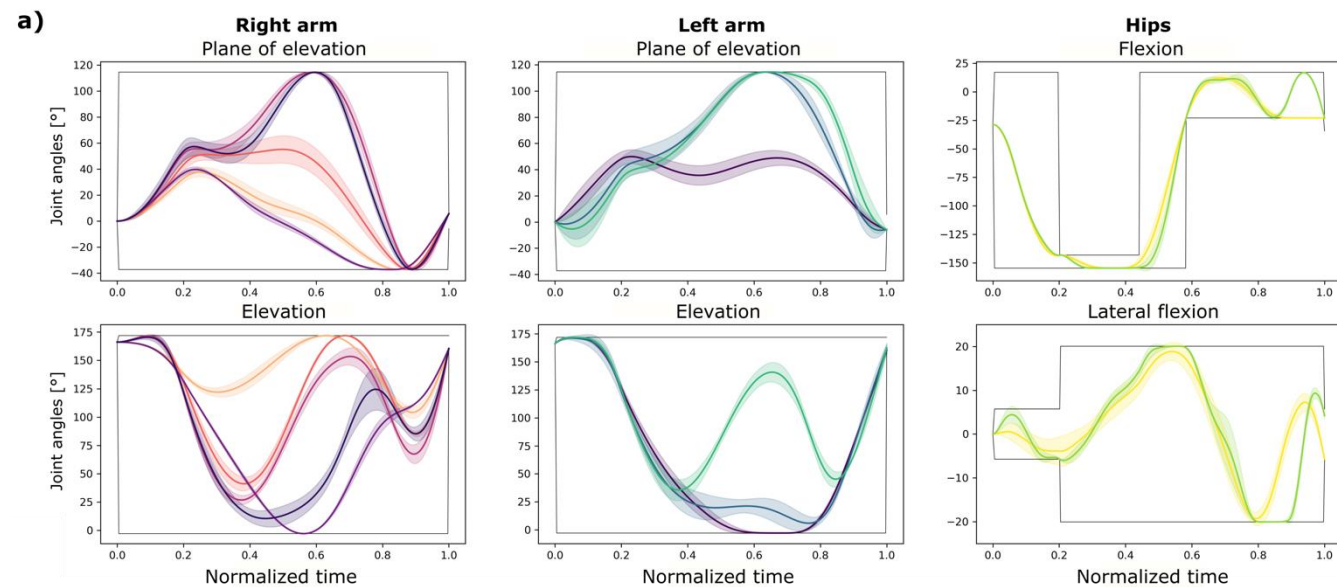
a)



Conclusion

⑥ Acrobatic techniques are not *"one size fits all"*

- Try to work with the athlete's first intuition before coaching the generic technique
- If the technique does not work out for this athlete, do not hesitate to try other ones! (you can make educated guesses based on twist potential and creativity)



4 Techniques are dependent on the athlete's anthropometry

Bonus

It is possible to make small anthropometry changes

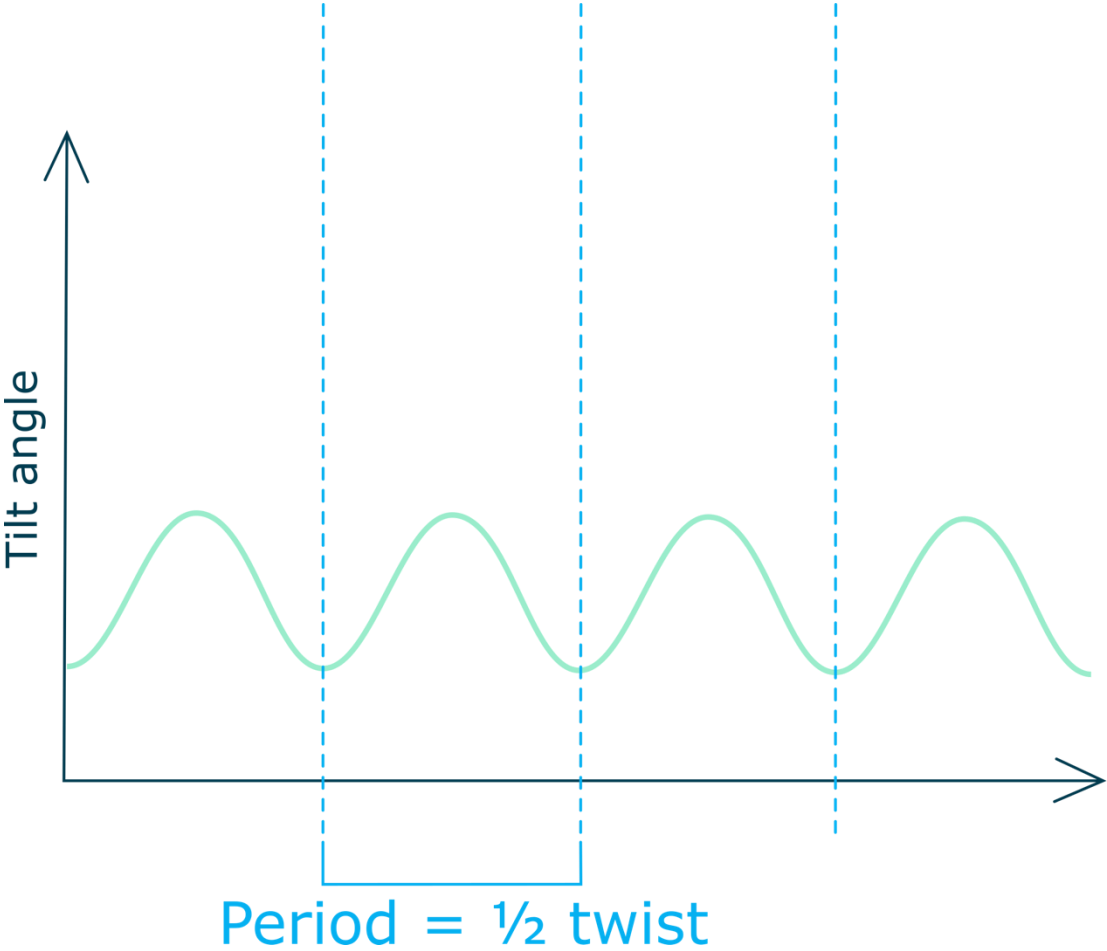
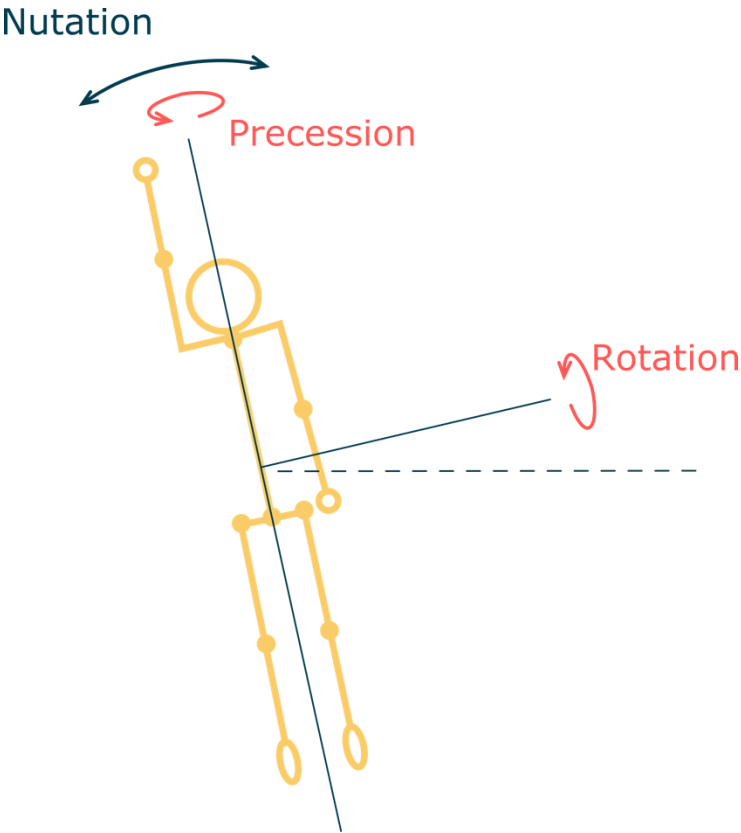
- Equipment
- Physical conditioning



- 1 Moving segments in the best tilting plane
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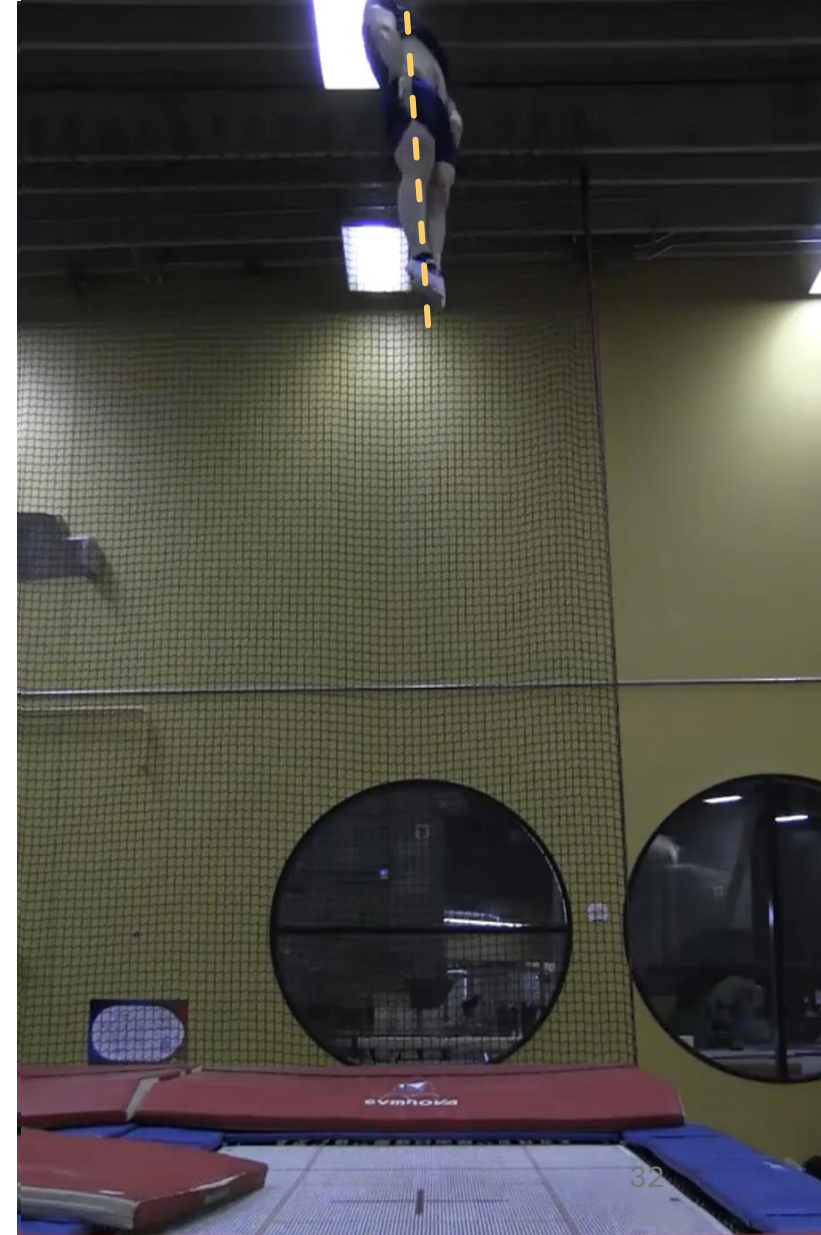
5

Moving segments at 1/4 twist timings



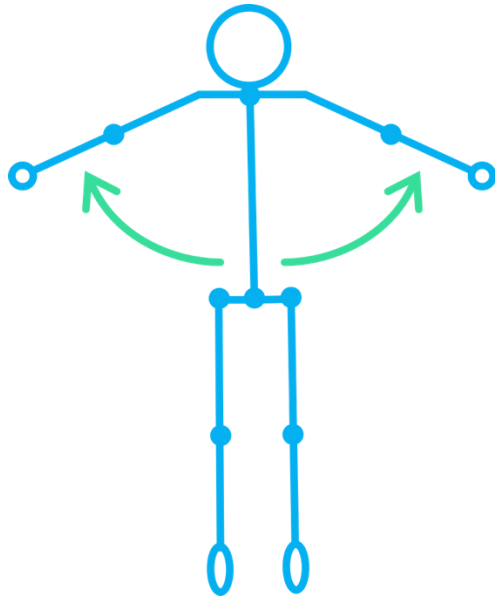
5

Moving segments at $\frac{1}{4}$ twist timings

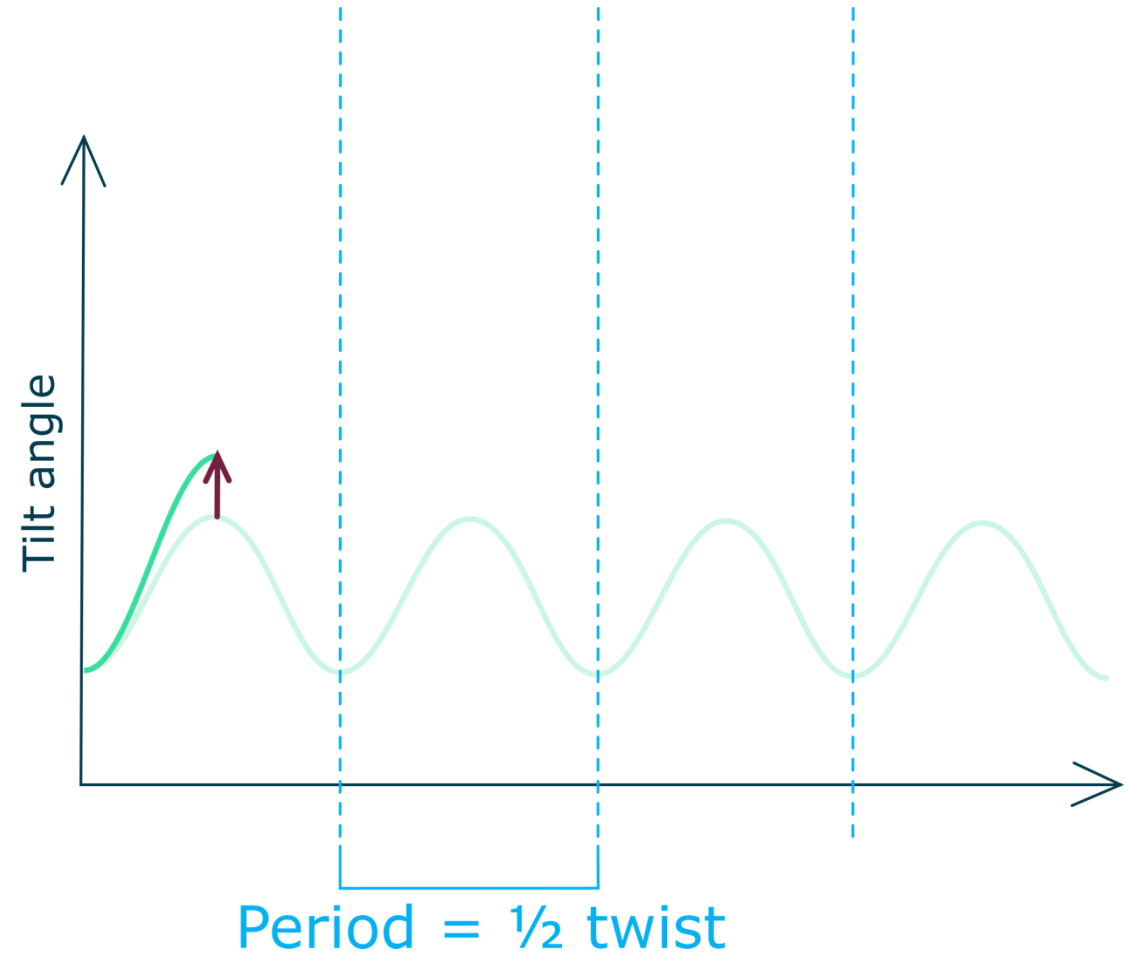


5

Moving segments at $\frac{1}{4}$ twist timings

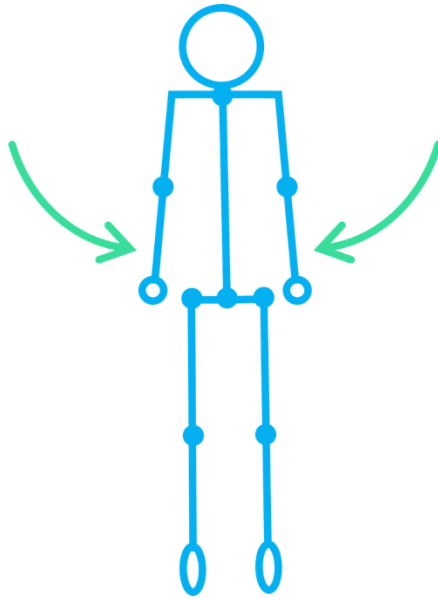


Each $\frac{1}{2}$ and full twist,
the arms can be abducted

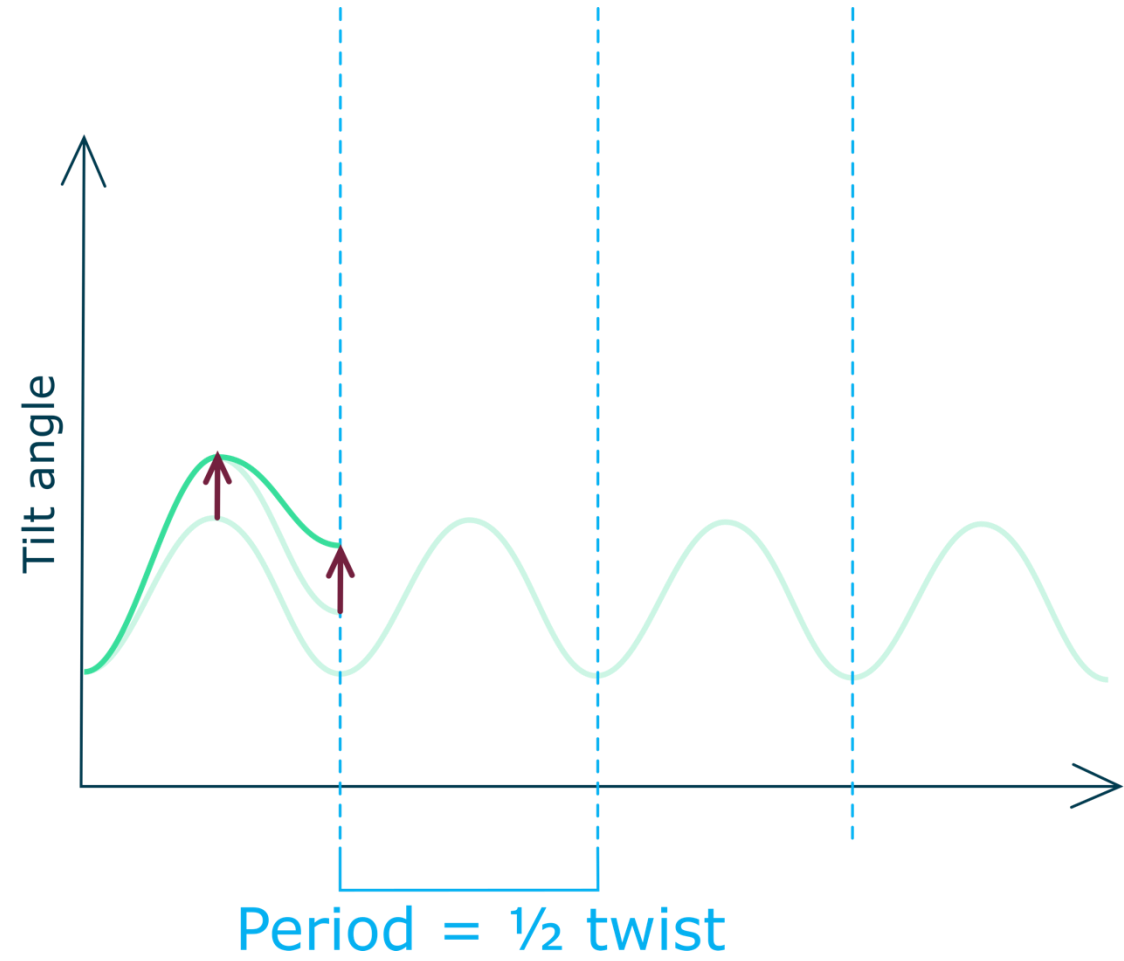


5

Moving segments at $\frac{1}{4}$ twist timings

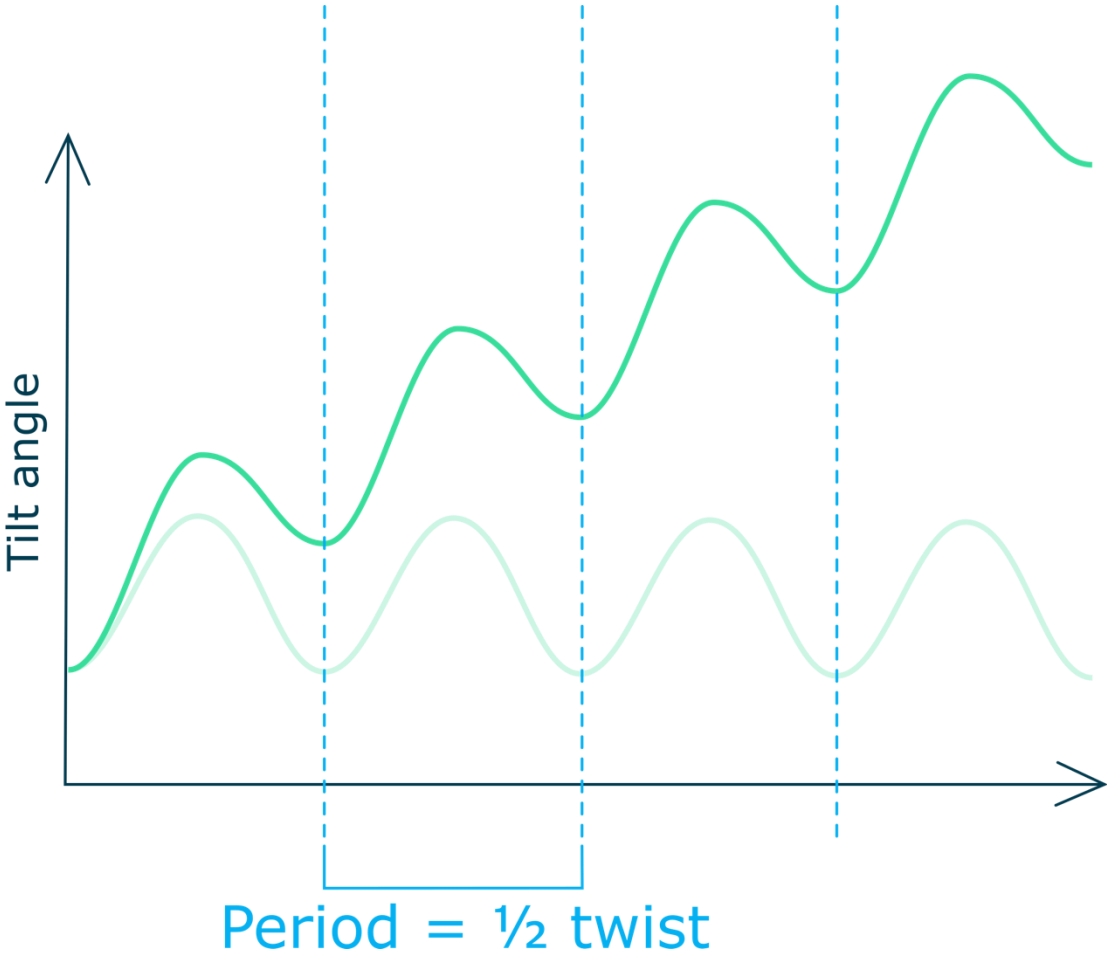
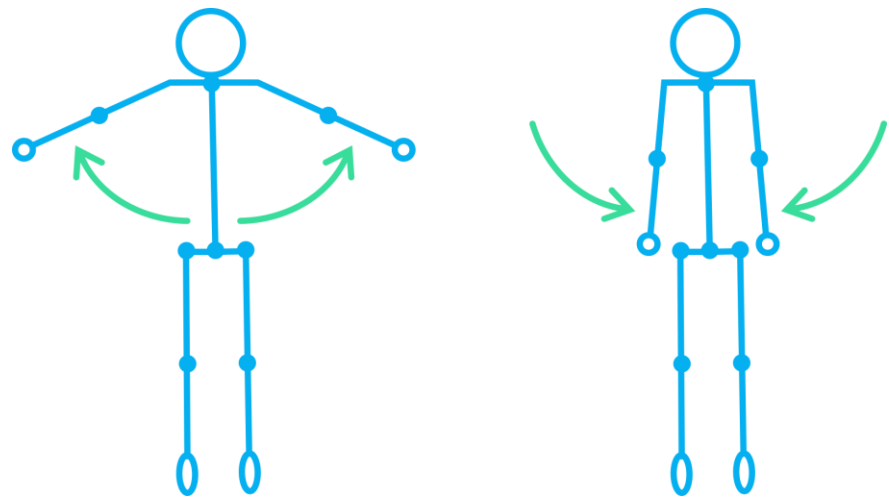


Each $\frac{1}{4}$ and $\frac{3}{4}$ twist,
arms can be adducted



5

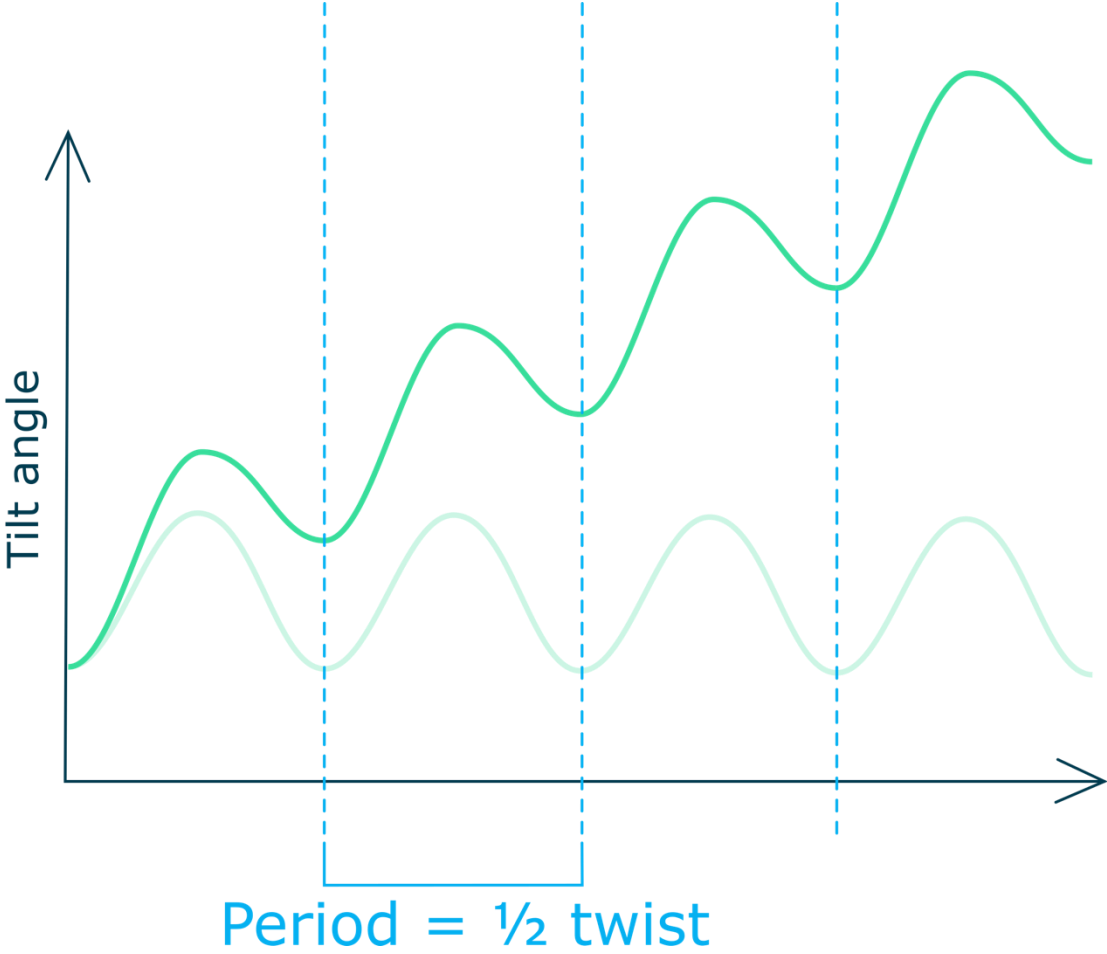
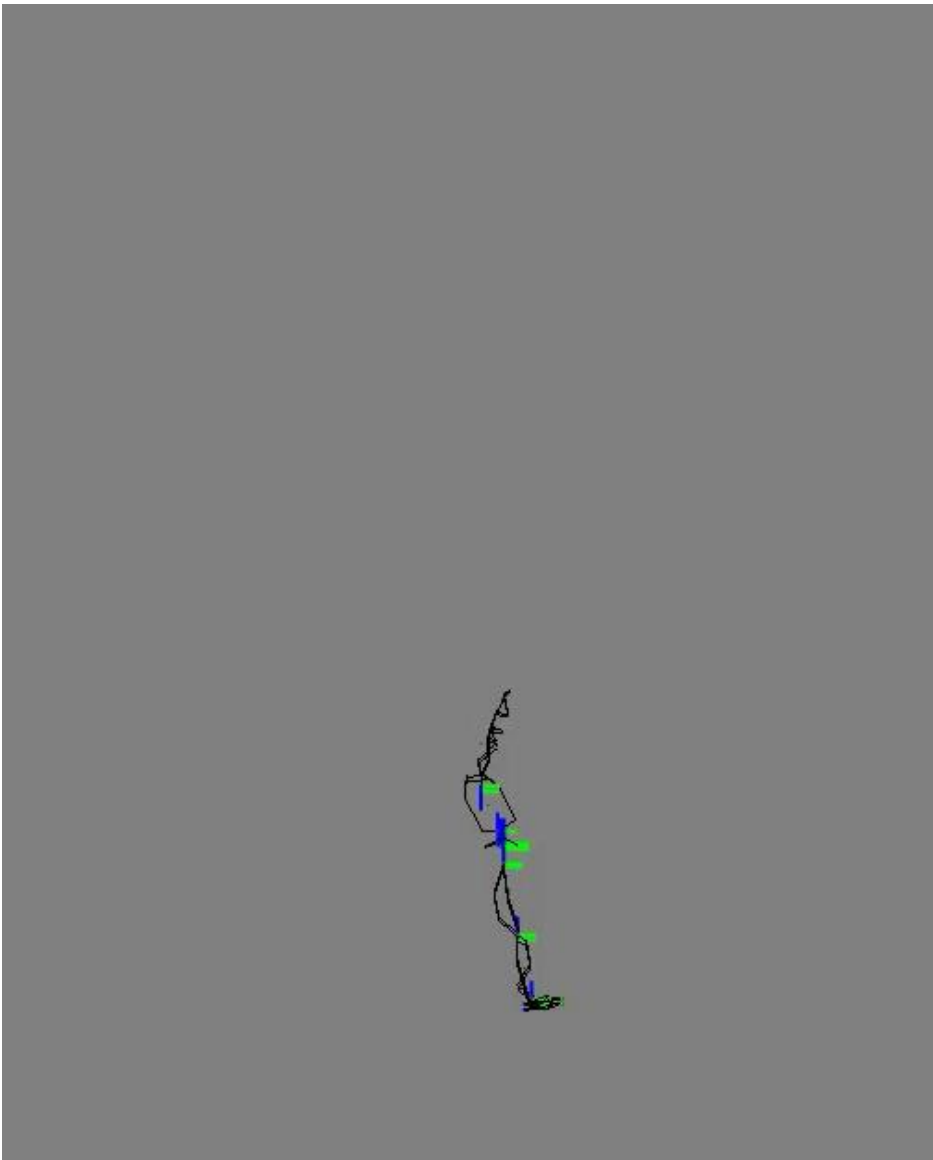
Moving segments at 1/4 twist timings



Even with symmetrical arm movements, it is possible to accelerate twist!

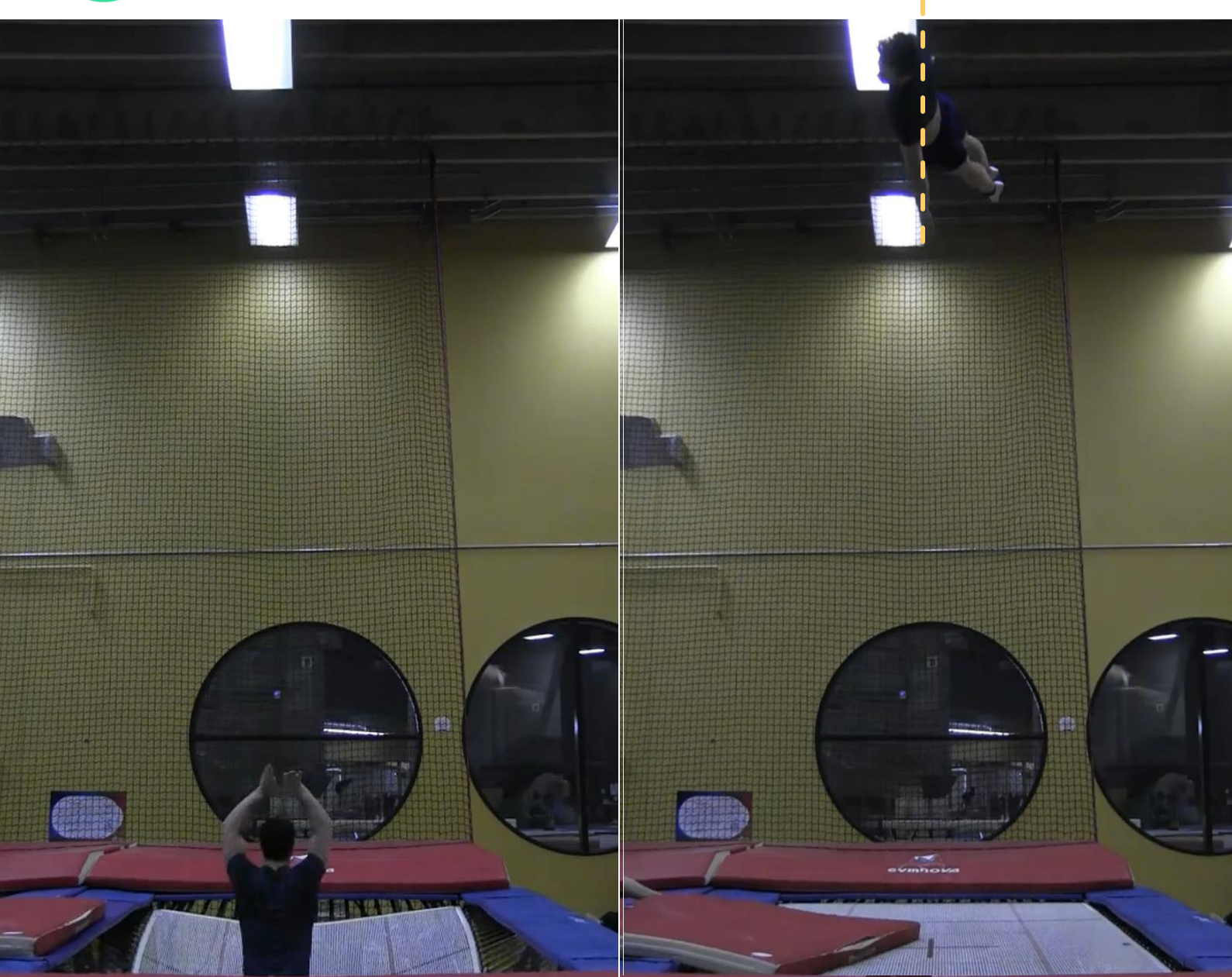
5

Moving segments at 1/4 twist timings



5

Moving segments at $\frac{1}{4}$ twist timings



Conclusion

- ⑦ Changes in inertia at $\frac{1}{4}$ twist timings are more efficient
- You can use nutation to accelerate twist (contact twist least visible)
 - Keep in mind that even symmetric movements can modify tilt angle

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6 Looking at the trampoline without fixating

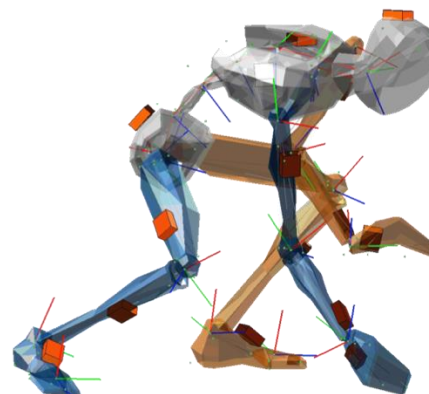


8 elites vs 9 subelites



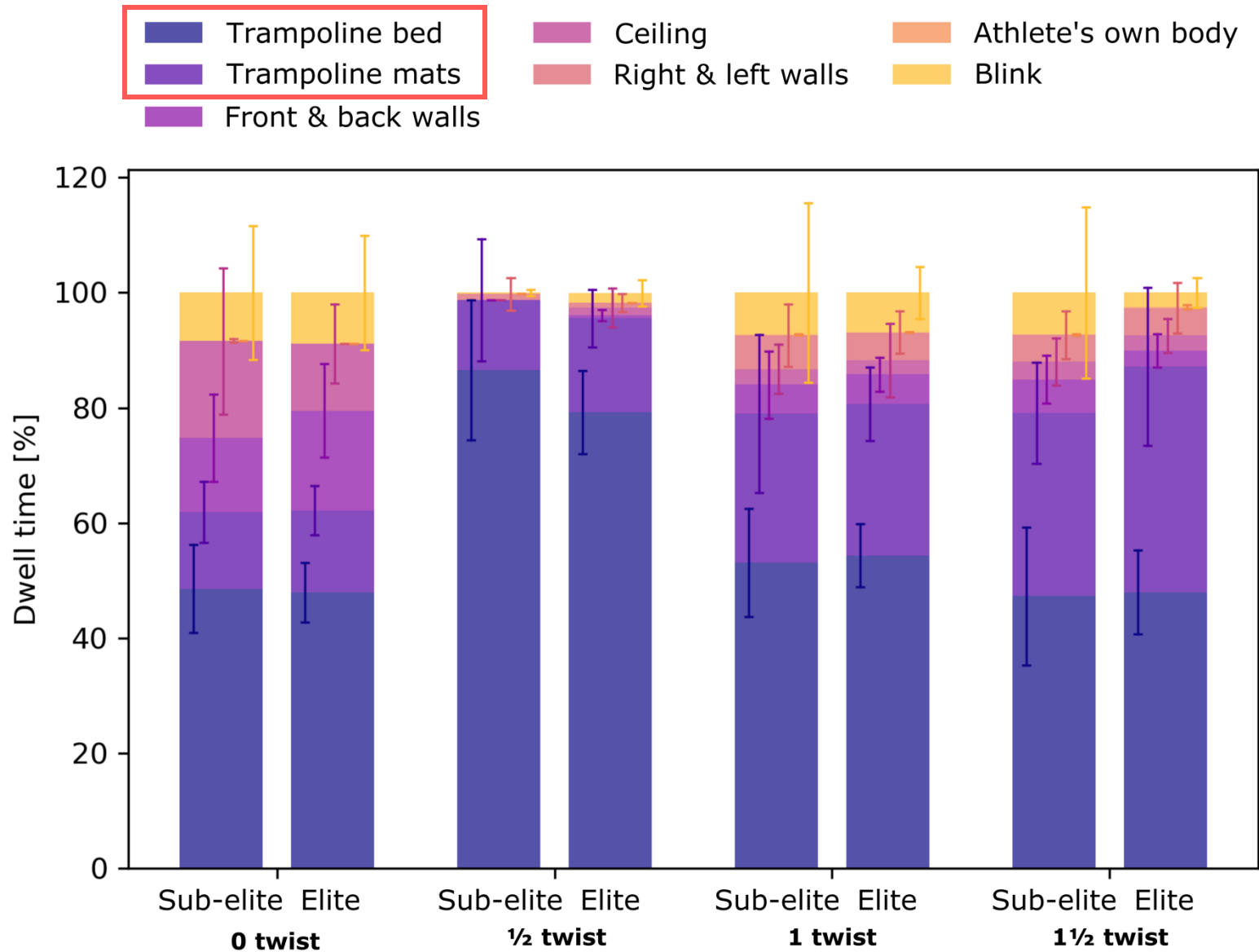
Pupil
Labs

xsens

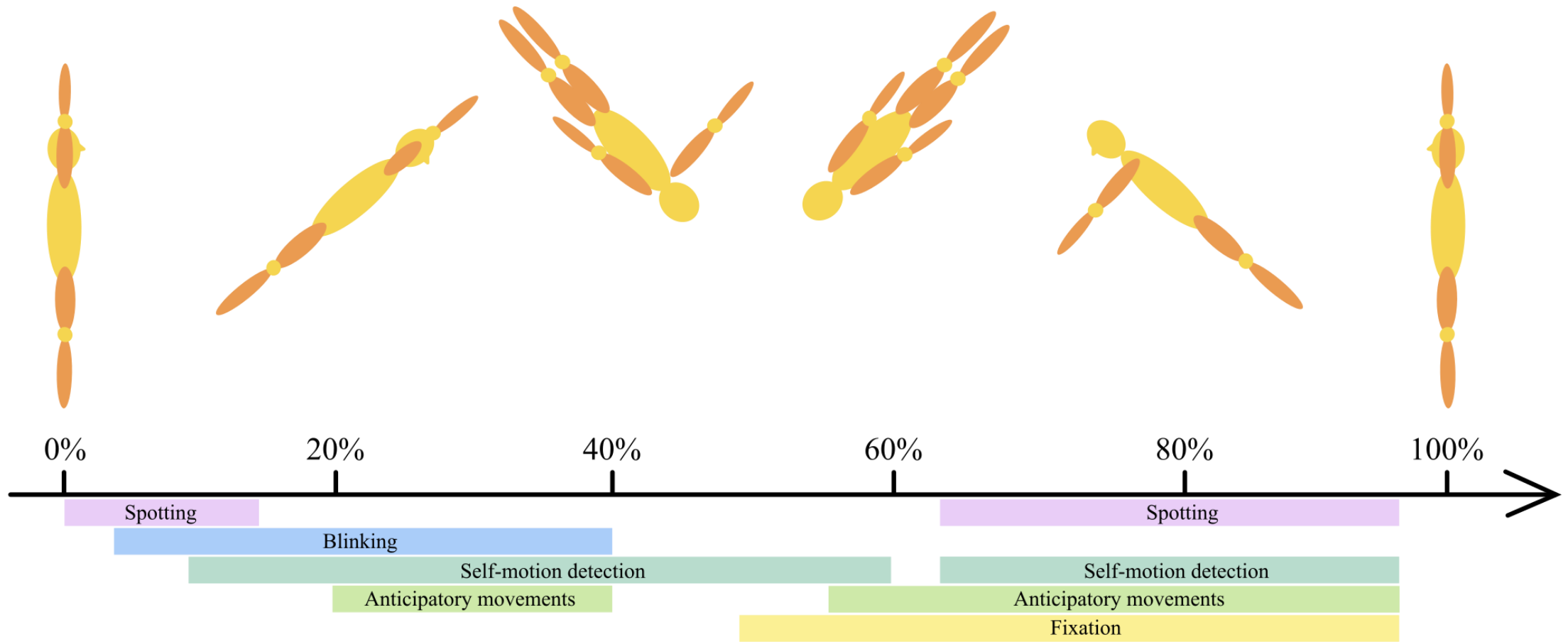


6

Looking at the trampoline without fixating



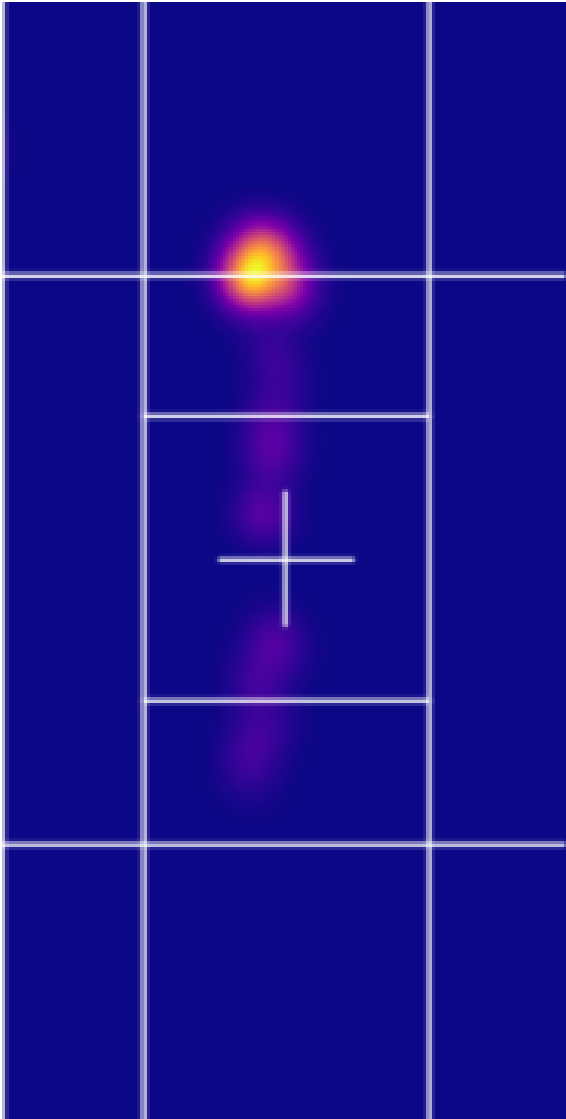
6 Looking at the trampoline without fixating



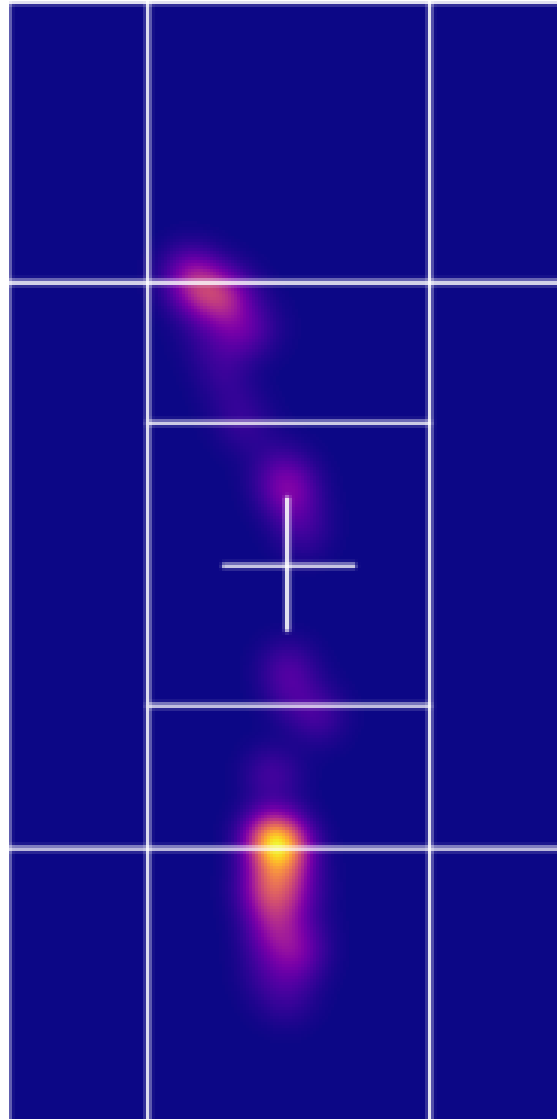
6

Looking at the trampoline without fixating

Continuous gaze



Fixations / saccades



Conclusion

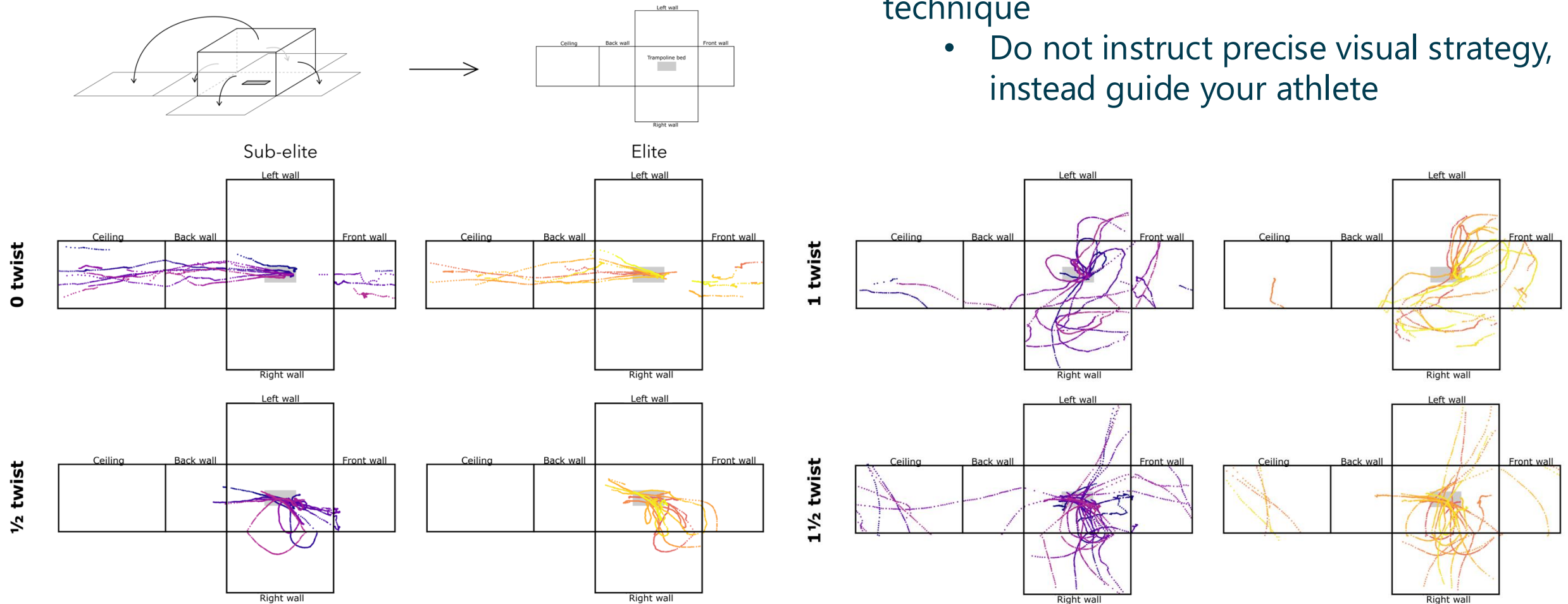
- ⑦ Peripheral vision might be more important than foveal vision for acrobatic control
- Do not ask athletes to fixate their gaze on specific visual cues
 - Do not encourage abrupt changes in visual orientation (makes athlete temporarily blind)

6

Looking at the trampoline without fixating

Conclusion

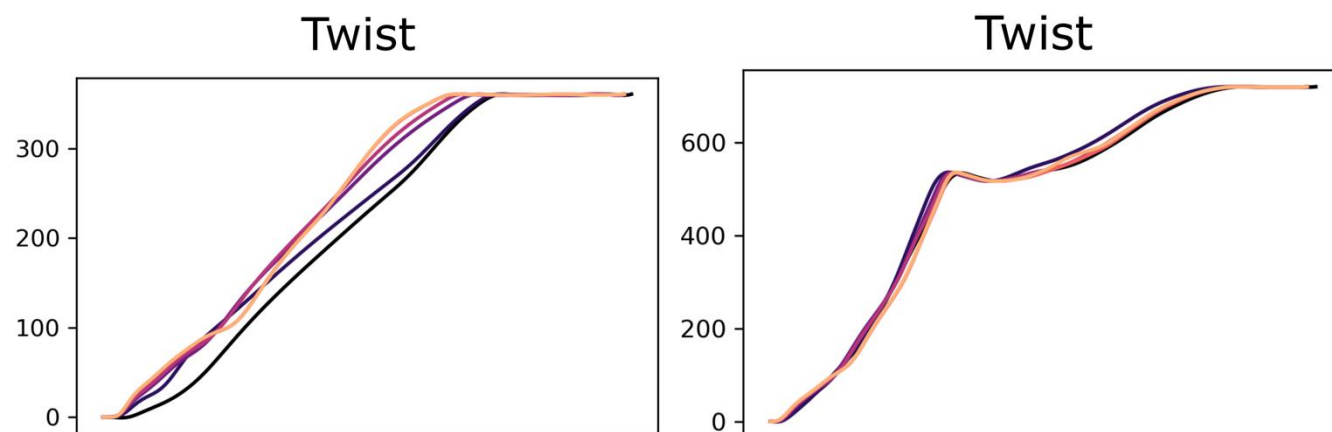
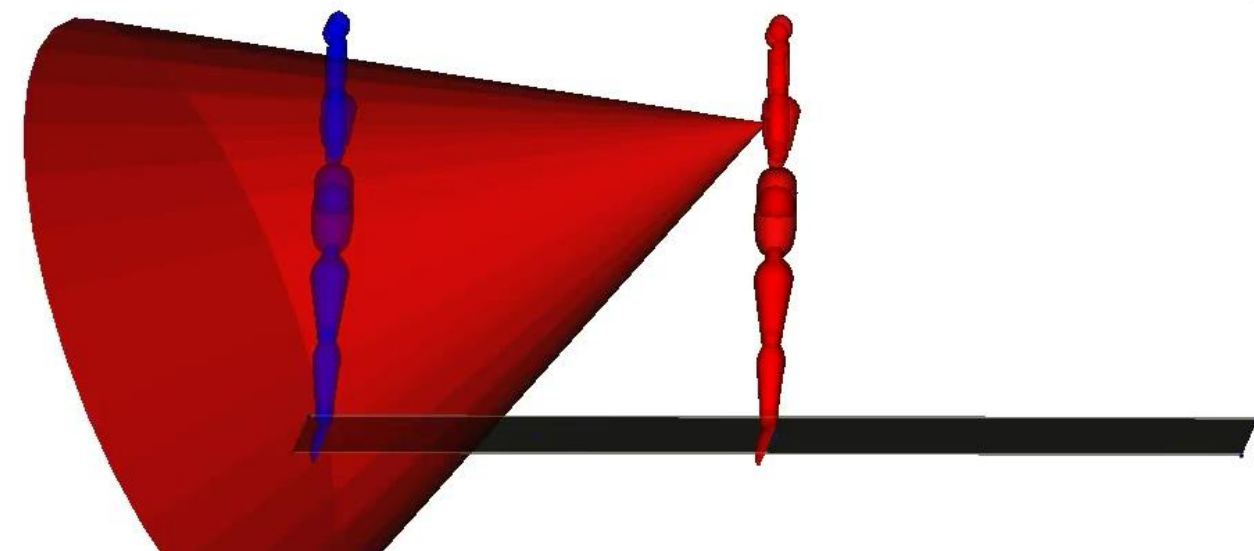
- ⑧ Large inter-athlete variability in visuomotor technique
- Do not instruct precise visual strategy, instead guide your athlete



- ① Moving segments in the best tilting plane
- ② Circular motion of the hips
- ③ Adjustments might be needed before piking
- ④ Techniques are dependent on the athlete's anthropometry
- ⑤ Moving segments at $\frac{1}{4}$ twist timings
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- ⑦ Vision helps regulating variability

7

Vision helps regulating variability



7

Vision helps regulating variability

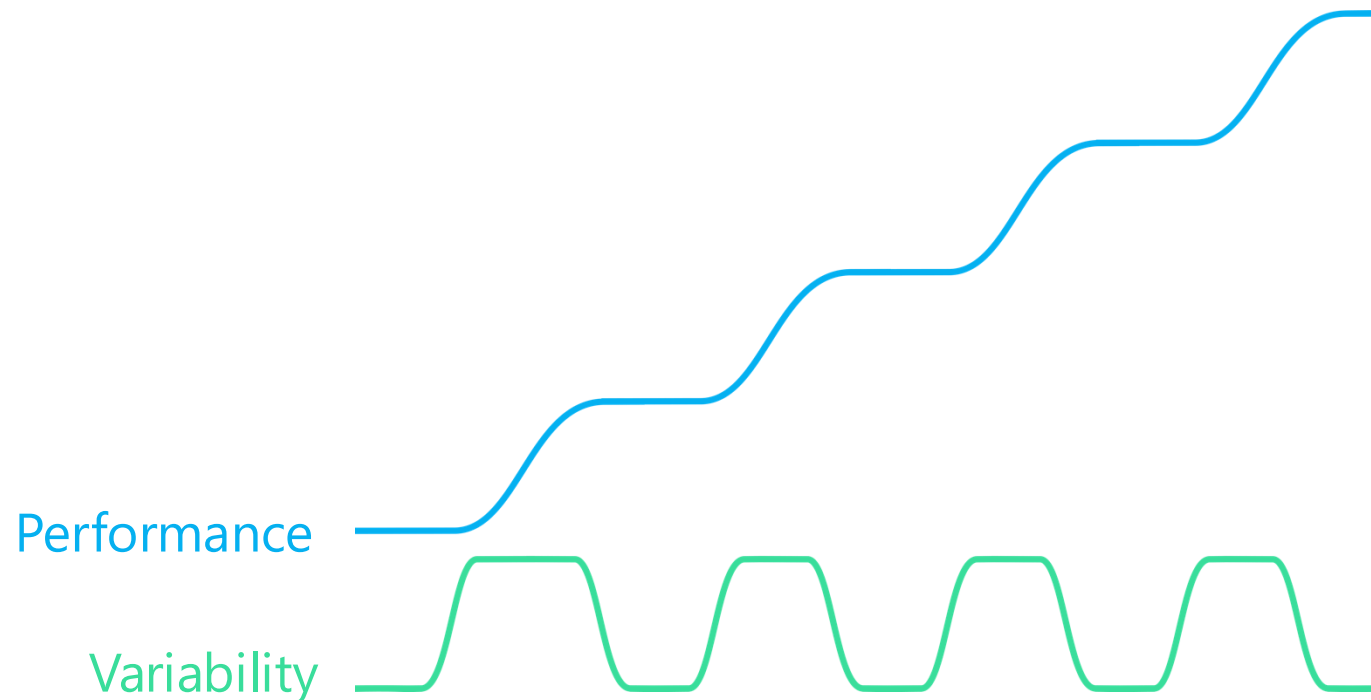


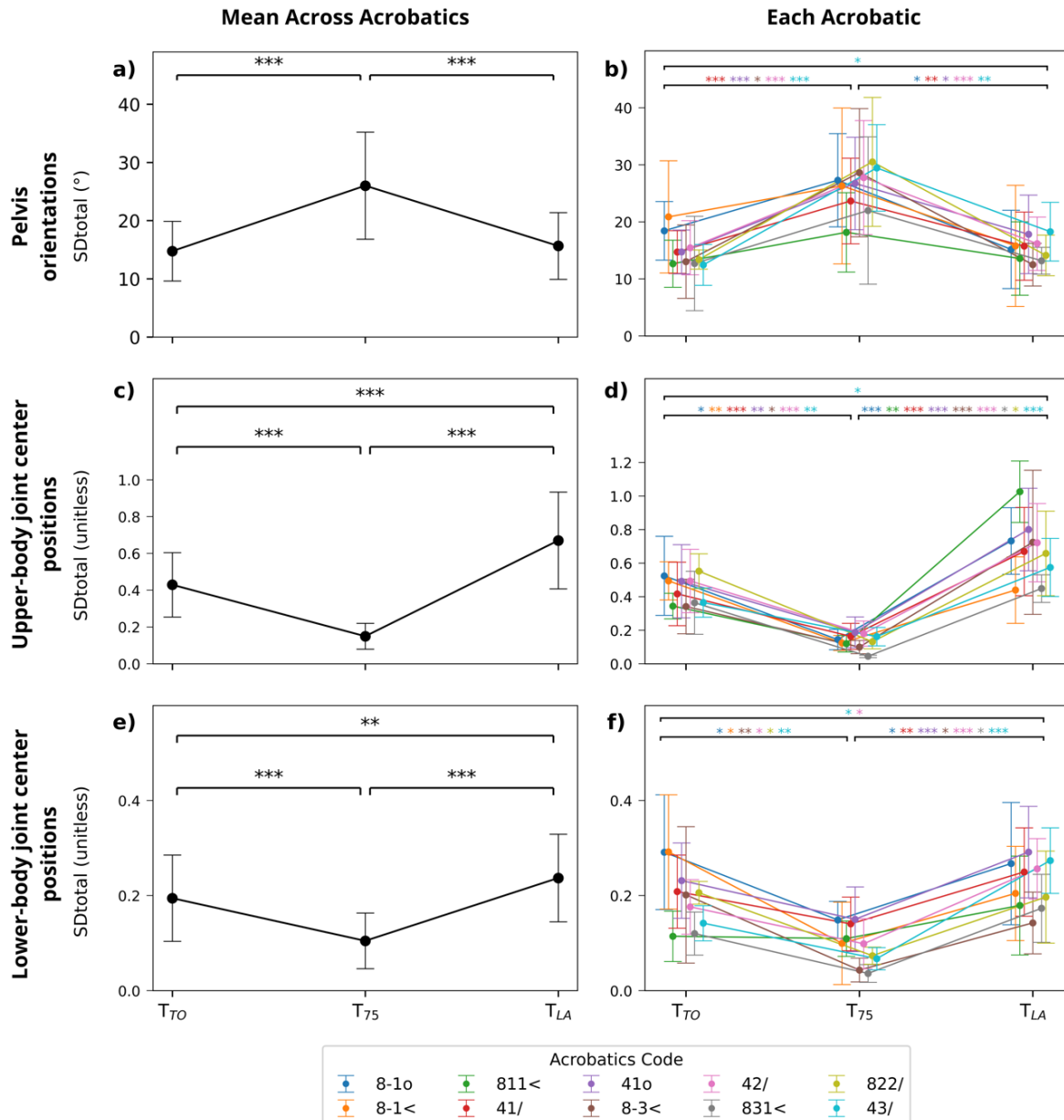
Conclusion

- ⑧ Consider visual and motor techniques together
- Vision can affect kinematics
 - Athletes might not be aware

Conclusion

- ⑨ Know your athletes
- Take advantage of their strengths
 - Work on their weaknesses
 - Alternate Performance/Training modes





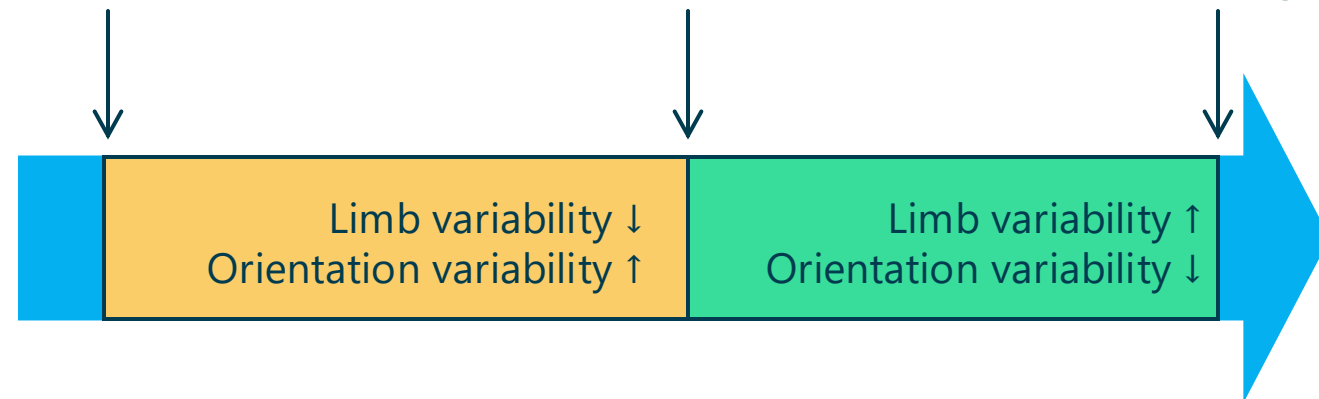
Conclusion

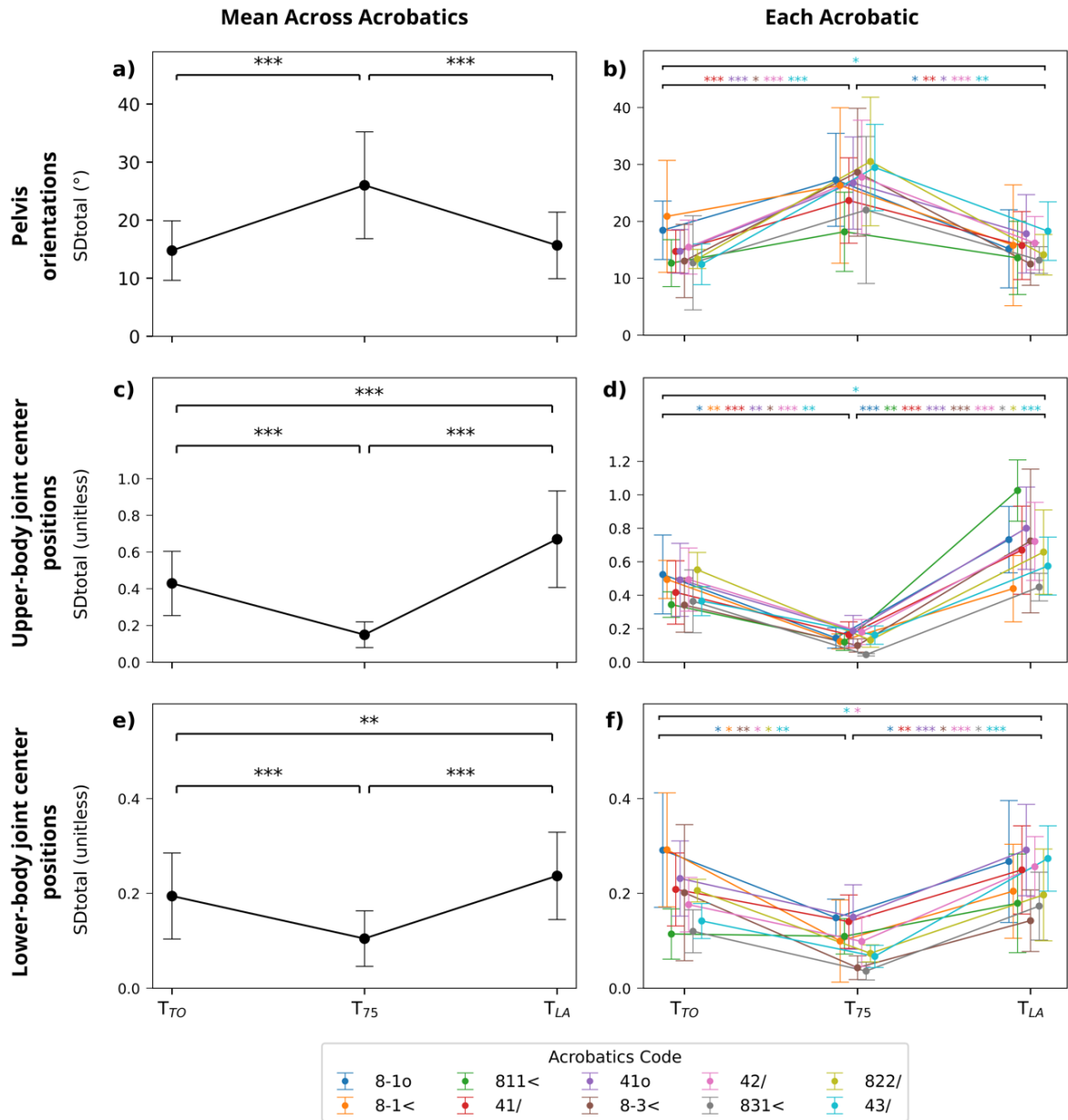
- ⑩ Anticipate adjustments before landing
- The technique must allow for different arm movements

Take-off

Visual info

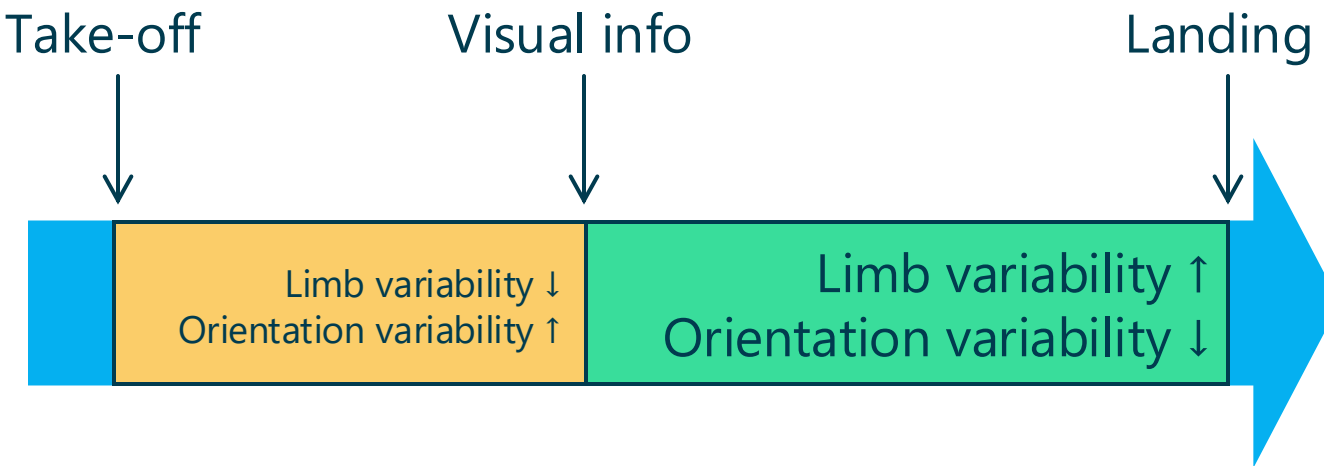
Landing





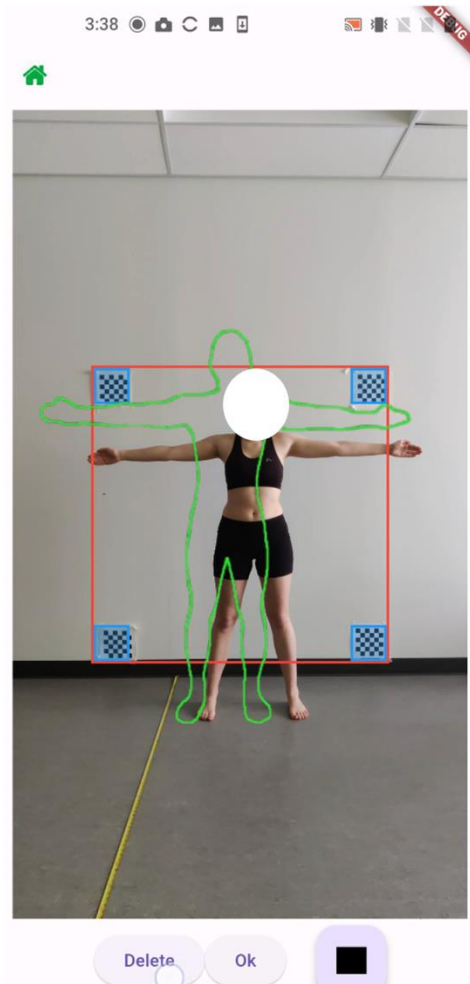
Conclusion

- ⑩ Anticipate adjustments before landing
- The technique must allow for different arm movements

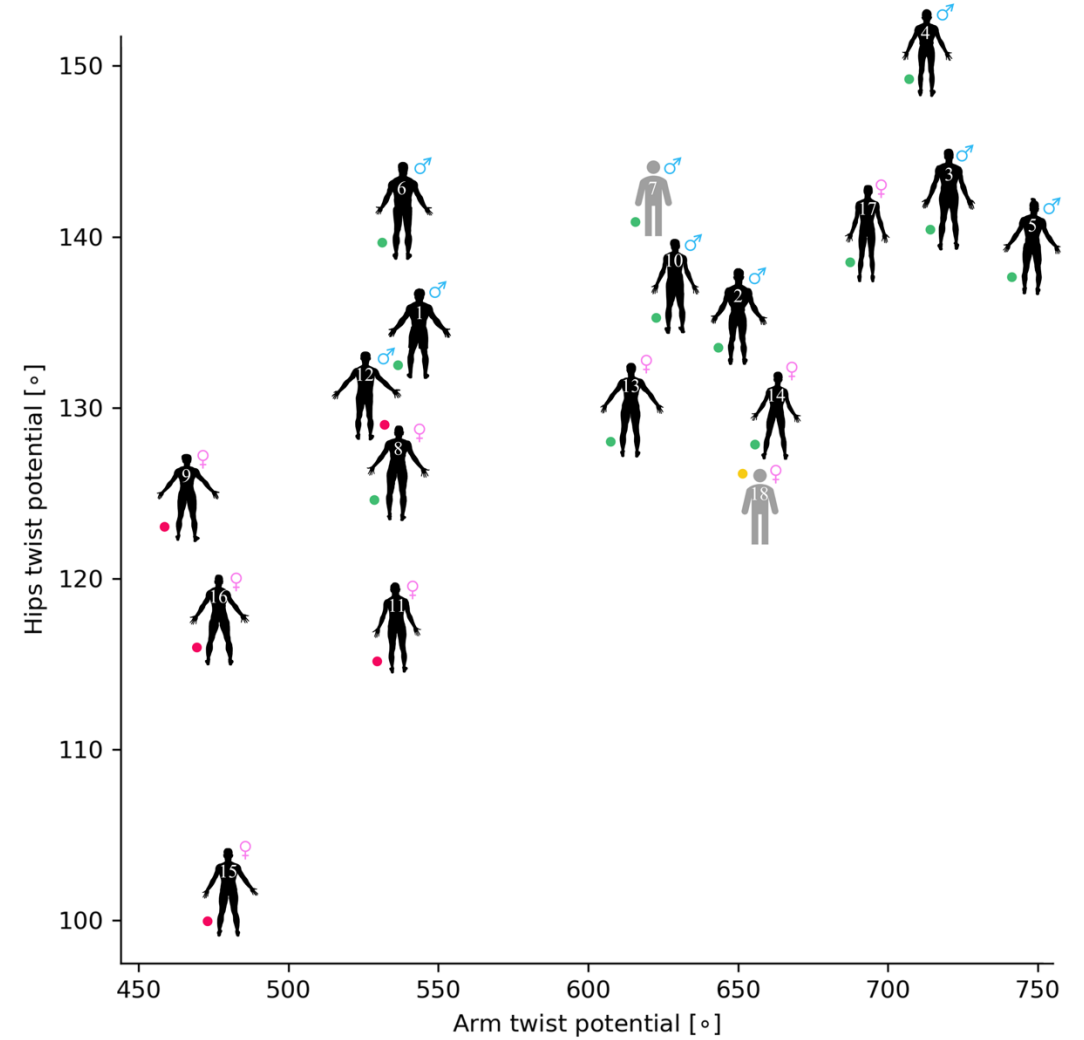
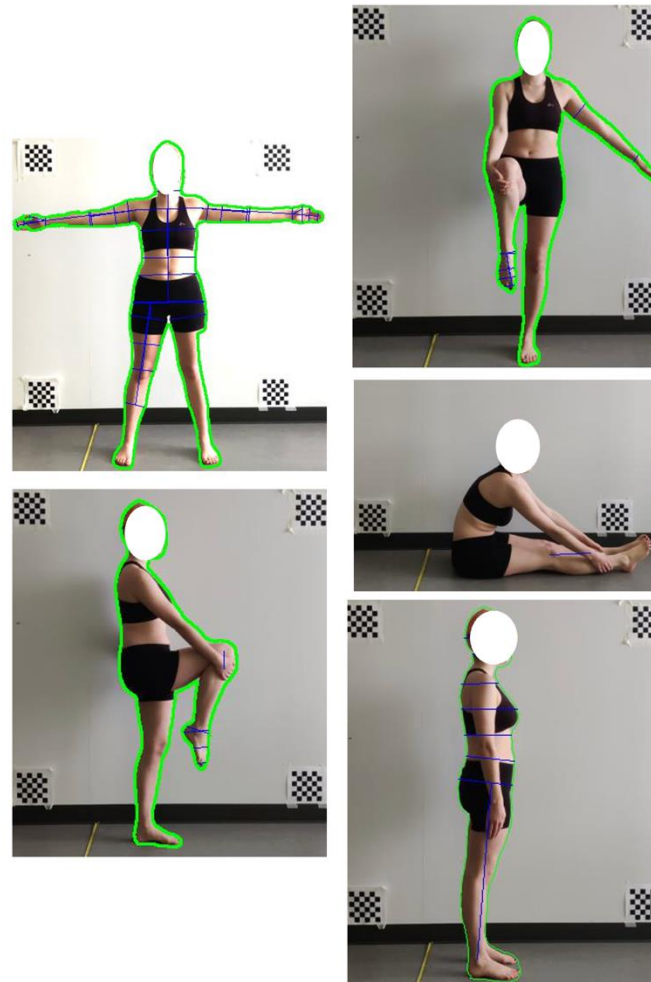


YeadonModelGenerator

a)



b)



5 pictures → Twist potential

BioptimGUI - Acrobatics

Optimal control type
Acrobatics optimal control program

Sport type
Trampoline

Dynamic equations
Torque driven

Model path *
Select the model file

Number of somersaults *
1

☐ Non-collision
☐ Visual Criteria
☐ With spine

Number of half twists *
0

Final time *
1.0

Final time margin *
0.1

Preferred twist side *
Left

Jump position *
Straight

Degrees of freedom

Information on Somersault 1

Number of shooting points
40

Phase time (s)
0.5

Objective functions

Constraints

Information on Landing

Number of shooting points
40

Phase time (s)
0.5

Objective functions

Constraints

Export script

Let's discuss!

- ① Moving segments in the best tilting plane
- ② Circular motion of the hips
- ③ Adjustments might be needed before piking
- ④ Techniques are dependent on athletes' anthropometry
- ⑤ Moving segments at $\frac{1}{4}$ twist timings
- ⑥ Looking at the trampoline without fixating
- ⑦ Vision helps regulating variability

